DEPARTMENT OF CYBER SECURITY

AIR FORCE INSTITUTE OF TECHNOLOGY, KADUNA

IMPLEMENTING BLOCKCHAIN FOR CERTIFICATE ISSUANCE AND AUTHENTICATION IN NIGERIAN UNIVERSITIES

BY

SUNDAY FRANK OCHIGBO

U20CYS1087

SEPTEMBER 2024

DECLARATION OF ORIGINALITY

Project Title: Implementing Blockchain for Certificate Issuance and Authentication in Nigerian Universities
Student Name: SUNDAY FRANK OCHIGBO
Main Supervisor: MR. TAJUDDEEN MUHAMMAD MASHKUR
Co-Supervisor(s):

This project report is submitted as part of the requirement for the Bachelor of Science in Cyber Security at the Air Force Institute of Technology. I confirm that the work contained in this B.Sc. project report has been solely composed by myself, except where indicated in the text, and has not been accepted in any previous application/award for a degree in any institution. The report may be freely copied and distributed provided the source is acknowledged.

____

Date:

CERTIFICATION OF APPROVAL

IMPLEMENTING BLOCKCHAIN FOR CERTIFICATE ISSUANCE AND AUTHENTICATION IN NIGERIAN UNIVERSITIES

by

SUNDAY FRANK OCHIGBO

U20CYS1087

Meets the requirements governing the award of the degree of **BACHELOR OF SCIENCE** in **CYBER SECURITY** and is hereby approved for its contribution to knowledge and literary presentation.

______________________________	______________________________
MR. TAJUDDEEN MUHAMMAD MASHKUR Project Supervisor	Date
______________________________	______________________________
MR. DAUDA SULE Project Coordinator	Date
______________________________	______________________________
SQN LDR SM ABDULLAHI Head of Department	Date
______________________________	______________________________
Prof. ISMAILA IDRIS External Examiner	Date

DEDICATION

In pursuit of knowledge, guided by curiosity, and fuelled by determination, I dedicate this research to the divine spark that resides within us all. May it illuminate the path of discovery, inspire wisdom, and bring forth understanding. To the Source of all existence, I offer my gratitude and reverence.

ACKNOWLEDGEMENTS

I am profoundly grateful for the steadfast support and guidance provided by the Department of Cyber Security at every stage of the research process. My heartfelt thanks also go to my parents, Mr. Ogoh Sunday and Mrs. Enyimoche Angela, whose unwavering love and encouragement have been the cornerstone of my journey. I extend my deepest appreciation to my supervisor, Mr. Tajuddeen Muhammad Mashkur, for his invaluable mentorship and consistent support. Furthermore, I am sincerely thankful for the collaborative efforts of my colleagues and friends, whose diverse perspectives and insights have significantly influenced the course and outcomes of this research project.

ABSTRACT

Issuing and verification of academic certifications such as degrees in Nigerian universities is a very cumbersome process due to the archaic procedures based on paper. They are expensive, inefficient, and hard to apply with high chances of being manipulated. A trustworthy, secure, and efficient system is extremely important to take care of these responsibilities. This project aims to change the way certificates are issued and verified in all Nigerian universities through blockchain technology. And that's where this solution merges with using the blockchain to create a platform for these universities that shall issue certificates and store them in a way that is immutable, and no one apart from the authorized bodies will be able to alter it. A full range of testing was carried out in the process to ensure scalability, safety, and reliability of the platform. The implementation of this blockchain solution will be a cornerstone for helping the certificate management operation flow better, reducing administrative workloads, protecting from academic certificate fraud, and improving trust in certification. It would also stimulate inventiveness and technological innovations in the educational sector, and it might be a step closer to another leapfrogging of Nigerian higher education. Generally speaking, this is a significant stride in ridding Nigerian universities off fraudulent, paper-based certificate management systems.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF ABBREVIATIONS

ABI	Application Binary Interface
API	Application Programming Interface
CSS	Cascading Style Sheets
DApp	Decentralized Application
DAO	Decentralized Autonomous Organization
EVM	Ethereum Virtual Machine
GIF	Graphics Interchange Format
HTML	Hypertext Markup Language
IDE	Integrated Development Environment
IPFS	Interplanetary File System
JCR	Journal of Critical Reviews
NUC	National University Commission
PDF	Portable Document Format
PNG	Portable Network Graphic
QR	Quick Response
SVG	Scalable Vector Graphics

CHAPTER ONE

1.1 INTRODUCTION

Although educational certificates validate academic achievements, many have been compromised due to rapid advances in multimedia and printing technology, which have facilitated the production of countless fake certifications (Smith, 2023). Counterfeiting methods may include creating new certificates, forging legitimate degrees, or using print shops. Verification is a cumbersome process and time-consuming; also, chances of fraud are very high in case the candidate has colluded with the issuing institution.

The technology applied here is blockchain technology; our certificates are safely preserved, and there is no need for a third party to verify them. The data is secured and preserved in a tamper-proof distributed ledger, which makes it quite secure and cost-reductive, as well as less risky when it comes to the maintenance of physical certificates. The broad visibility and trustworthiness of blockchain technology bring an optimal solution for the validation of academic certificates to block fraud (Geetha S K, 2020).

1.2 BACKGROUND TO THE STUDY

There's no doubt that technology has significantly simplified and enhanced our daily lives. However, it's also created a significant challenge for Nigerian universities. The rise in fraudulent academic certificates being used to secure jobs is now a major concern. JAMB Registrar Is-haq Oloyede disclosed that 815 A-Level certificates, 453 university diplomas, and 397 College of Education certificates were found to be fraudulent (Punch Newspaper, 2024). With global population growth and resources becoming scarce, competition for jobs has increased, highlighting the increased need to tackle the issue. In 2019, The PUNCH Newspaper reported that Nnamdi Azikiwe University, Awka, dismissed a lecturer due to holding a fraudulent master's degree, committing plagiarism, and being employed at multiple institutions concurrently. The lecturer was also accused of possessing a fake

BSc degree from a university in Rivers State. Fortunately, blockchain technology offers a promising solution for Nigerian universities.

Blockchain technology is a distributed digital ledger enabling secure and transparent transactions without intermediaries (Techopedia, 2024). Since its introduction by Satoshi Nakamoto, it has unlocked numerous possibilities, including certificate verification. Adopting a more robust platform for verifying academic certificates using blockchain technology can enable Nigerian universities to hire only qualified candidates, thereby fostering a fair and equitable society. Blockchain technology not only improves the verification process but also offers substantial benefits in cybersecurity—a critical concern in today's digital age. Universities, as custodians of sensitive data, must ensure that this information is securely stored and accessible only to authorized individuals. The decentralized nature of blockchain technology provides a solution to this by ensuring that data is not governed by a single entity, which significantly reduces the risk of cybercriminals hacking into the system and tampering with or stealing the data.

This research intends to review previous studies on academic certificate verification in Nigerian universities and explore how blockchain technology can be utilized to enhance cybersecurity. Blockchain provides a secure and tamper-resistant method for storing and validating academic credentials. By integrating this technology, Nigerian universities can reap substantial benefits from the increased security and transparency it offers, while also ensuring their data is well-protected against cyber threats.

The implementation of blockchain technology in Nigerian universities will not only curb academic fraud but also ensure that only eligible candidates are employed and that sensitive data remains safeguarded from cyber-attacks. Such advancements will pave the way for a brighter future for the country, contributing to the establishment of a secure and just society.

1.3 PROBLEM STATEMENT

In Nigeria, it's common for individuals to earn degree certificates from their universities, which they use to prove their qualifications for specific courses. Employers often ask for these certificates when hiring. Unfortunately, some job applicants attempt to mislead employers by submitting fraudulent certificates during the hiring process. If these documents are not properly verified, there is a risk of hiring unqualified candidates, which can lead to various issues within the organization. These problems may include diminished performance, reduced productivity, and potential legal consequences.

To prevent such situations, it is crucial for employers to verify the authenticity of the certificates provided by candidates. One common method is to reach out to the university where the candidate claims to have earned their degree and confirm the validity of the certificate. However, this process can be time-consuming and may vary in efficiency depending on the responsiveness of the university being contacted. Additionally, candidates may collude with the university to falsify their certificates, making it difficult to detect fake certificates (Punch Newspaper, 2024). To address this issue, the implementation of blockchain-based smart academic certificates could be a viable solution that is tamper-proof and easy to verify in the Nigerian society. With smart academic certificates, employers can easily and quickly verify the authenticity of a candidate's certification, reducing the chances of hiring unqualified individuals and ultimately improving the quality of the labor force (Geetha, 2020).

1.4 AIM AND OBJECTIVES

The goal of this research is to create a secure, decentralized blockchain platform for the issuance and authentication of certificates in Nigerian universities.

The objectives focus on using the benefits of blockchain technology to solve particular problems related to managing and checking academic certificates in Nigerian universities, making the academic processes more trustworthy and secure. The primary objectives for the implementation of blockchain technology in Nigerian universities are as follows:

I. **To Establish a Secure Platform for Certificate Issuance**: Design a decentralized blockchain platform that guarantees the secure issuance and authentication of academic certificates. This platform will minimize the risk of fraud and bolster trust in the credentials provided by universities.

II. **To Facilitate Certificate Hashing Processes**: The objective of this task is to automate the process of creating unique digital fingerprints (hashes) for certificates. With the use of blockchain technologies, secure techniques will be applied to make the certificates tamper-proof, easy to verify, thus building security and trust.

III. **To Verify Issued Certificates**: Streamline the verification process for academic certificates, making it quicker and more efficient by eliminating the need for manual checks and reducing dependence on third-party verification services.

1.5 SIGNIFICANCE OF THE STUDY

The integration of blockchain technology for issuing and validating certificates in Nigerian universities marks a ground-breaking shift with wide-reaching implications for the education sector. This approach aims to eliminate common issues like certificate fraud and forgery by leveraging blockchain's robust security features, ensuring that academic credentials are both authentic and tamper-proof (Effiong, 2020). The technology explored in this study simplifies the verification process and reduces administrative burdens by enabling the instant validation of academic qualifications. By leveraging blockchain, a transparent and trustworthy environment is created, which enhances the global reputation of Nigerian educational institutions (Omijeh, 2023). Additionally, digitizing certificates and storing them on the blockchain ensures the permanent and secure preservation of academic records, safeguarding students' achievements indefinitely.

With a reliable and easily accessible verification system, students and graduates can pursue further education and employment opportunities both locally and internationally, thereby promoting personal and professional growth. This study aligns with Nigeria's national strategy for technological advancement, positioning the country as a global leader in the application of blockchain technology within the education sector (Effiong, 2020). Moreover, the successful implementation of blockchain in this context could serve as a model for other sectors that require secure and verifiable digital data, such as healthcare records, property registration, and supply chain management.

This research makes a significant contribution to improving the security, efficiency, and integrity of academic credentialing in Nigeria, laying the groundwork for a more secure, transparent, and effective educational system (Geetha, 2020).

1.6 SCOPE AND LIMITATIONS

The research focuses on creating, developing, and deploying a blockchain-based system for certificate issuing and authentication in Nigerian universities. It examines the platform's architectural design, outlining the components and interactions required for safe certificate issuance and authentication. This research concentrates on developing methods for obtaining digital certificates, integrating them with existing university systems, and designing user interfaces for the Nigerian University Commission (NUC) and university administrative staff. It looks at ways for checking certificate authenticity, ensuring instant verification by stakeholders including employers and academic institutions. The research covers the identification and implementation of security protocols to safeguard the blockchain network, as well as the definition of user roles and permissions to assure permitted access. It guarantees that all necessary national standards are followed, such as the Nigerian data privacy legislation and educational requirements. The scope includes a trial deployment at a chosen university to assess functionality, usability, and effectiveness, with input relied on to improve. It measures the system's performance in terms of security, efficiency, and user happiness, as well as the general influence on certificate authenticity and university reputation in Nigeria. The research investigates the system's scalability to handle many institutions and discusses potential future additions, such as incorporating more academic records and foreign verification systems.

1.6.1 LIMITATIONS

Nigerian universities can benefit greatly from deploying a blockchain-based system for issuing and verifying certificates, however, this would come with some problems too:

I. **High Gas Prices:** The possibility of high gas prices on the Polygon network (especially when there is a lot of demand) might be taxing financially to the Nigerian Education Board (NUC) and universities.

II. **Technological Difficulty:** The first technological barrier takes the investment of a whole layer 3 infrastructure and support from university staff.

III. **Integration Challenges:** It will take time and expense to connect the blockchain system with not only existing university databases but also administrative processes.

IV. **Internet and Digital Literacy:** Places with poor internet connection or low digital literacy may cause some difficulties in implementing it effectively.

V. **Upkeep and Maintenance:** Although blockchain technology is inherently secure and transparent, it is not entirely immune to hacking, which makes regular monitoring and checks essential to ensure the integrity and robustness of the blockchain.

While a blockchain-based certificate system offers significant advantages, it is crucial to address the potential limitations mentioned above carefully. Sustaining such a system will require meticulous attention to various factors to ensure its long-term effectiveness and security.

1.7 ORGANISATION OF THE STUDY

This research work is divided into five (5) chapters and references section.

Chapter one is the introduction and chapter gives background to the study, statement of the problem, objectives of the study, Significance and scope of study.

Chapter two is literature review and theoretical framework. Methodology is laid down in chapter three whereby research design is elaborated.

Chapter four deals with development and discussion of findings.

The final chapter of the study is chapter five, which contains the summation of observations in the study, conclusions made, contribution of the study to the existing body of knowledge, the limitations of the study, and finally, suggestions for future research.

CHAPTER TWO

LITERATURE REVIEW

It is crucial to have an understanding of related studies when conducting research in order to establish the foundation for further investigations. This understanding helps to gain knowledge about what has already been achieved and to identify the knowledge gap that the current research aims to fill. This chapter is intended to present background studies and to further explore reviews that are directly related to the objectives of this paper. This background study is sectioned into two parts. In Section 2.2, the initial part reviews previous studies relevant to this research, while Section 2.1 focuses on discussing blockchain technology.

2.1 CERTIFICATE SYSTEMS

There are generally two types of certification systems in today's educational landscape: there are several types of certificate systems that could be classified as non-blockchain, including: paper-based certificates, and; traditional digital certificates. As it is universally known, there are pros and cons to every system. The conventional system may contain paper-based security features, nonetheless, the current non-blockchain digital climate can be helpful since it allows for digital issuing, storage as well as validation of certificates. This is important because educational institutions are still discovering innovative methodologies as a means of improving their security and efficiency of their conventional certification through the use of block chain technology.

2.1.1 MANUAL PAPER-BASED CERTIFICATE SYSTEMS

A manual way of issuing academic certificates was the manual paper-based certificate system. Although this manual method will help enable printed documents on paper protected by one or more security features that are used to deter fraudulent behaviour, such as watermarking, holograms, sealing, hologram embossment stamp, and printing with heavy ink coverage. This method has been used in almost all educational institutions around the world, and Nigerian universities are not an exception.

Process

I. **Certificate Creation:**
   i. It starts with designing the certificate - featuring the university logo, name of graduate, degree awarded, date graduated, and addressing other necessary info.
   ii. They are developed using a method of printing that incorporation of various measures that enhance the stringency of the certificates to forging.

II. **Printing & Distribution:**
   i. Certificates are printed on high-security grade paper.
   ii. These certificates are presented to the graduates during the occasions such as graduations or through posts if the graduates cannot attend the occasions.

III. **Verification:**
   i. A university can authenticate a graduate's certificate when the graduate presents it to an employer or any other institution.
   ii. Records of all certificates issued can be found in the university and may be confirmed upon request.

IV. **Storage and Retrieval:**
   i. All graduates are responsible for keeping their physical certificates safe.
   ii. The universities also keep records of all the certificates they issue in their archives, to be verified from time to time.

V. **Authentication:**
   i. Public and private sector institutions may request an original certificate from graduates as a matter of procedure.

ii. Verification involves manually checking the certificate against the university's records.

Advantages

a. **Known Quantity**: This is a known system that all stakeholders are familiar with including universities, alumni, employers, and regulators.
b. **Physical Evidence:** Perhaps the most tangible benefit of owning a certificate is that you get real, physical possession over it which generally makes graduates feel proud and secure.
c. **Security Features:** Traditional security features such as watermarks, holograms, and embossing can help deter against simple attempts at forgery.

Disadvantages

a. **Fraud & Forgery:** Paper certificates are subject to complex frauds despite their many security features.
b. **Verification Delays:** Verification takes time and is a costly, cumbersome administrative process.
c. **Storage Issues:** Physical certificates can be lost or destroyed in addition to replacement of physical certificates taking time and process.
d. **Limited Accessibility:** Requires a physical presentation to show or verify the certificate, which makes this process less convenient for remote verification.

2.1.2 NON-BLOCKCHAIN DIGITAL CERTIFICATE SYSTEMS

Digital non-blockchain certificate systems utilize digital technology to issue, store, and verify certificates. These systems typically use digital signatures to encode electronic documents (usually PDFs) into centralized databases under the management of an issuing institution. This is because the aim is to make certification as fast and efficient as possible and include more layers of security- all with the click of a button.

Process

I. **Digital Issuance:**
   i. Certificates are issued in digital form (usually as a PDF) having digital signatures on them which assure that the certificate is real and not fake.
   ii. This certification is produced by the universities on a digital platform and then e-mailed to graduates, or they are downloaded from secure university portals.

II. **Storage:**
   i. They store all digital certificates in centralized databases that are under their control. To avoid data loss, these databases are backed up frequently.
   ii. Alternatively, graduates can store a digital copy on their personal devices or cloud storage services.

III. **Verification:**
   i. It provides online verification of a certificate which can be cross-checked through any employer or other interested parties using the centralized database.
   ii. This could require entering a certificate ID or following a secure link provided to you by the graduate.

IV. **Authentication:**
   i. Digital signatures and other cryptographic techniques are used to carry out the verification of the authenticity and integrity of a certificate.
   ii. This is quite often based on public key infrastructure (PKI), validated by the recipient (with their access to a trusted authority's PK) using the issuer's public key.

Advantages

a. **Convenience:** Digital certificates are issued and stored, sent digitally instead of through physical mail.

b. **Enhanced Security:** Digital certificates are regarded to be more secure in form of digital signatures and encryption than paper certificates.

c. **Accessibility:** All certificates will be digitally generated and can easily accessible from anywhere with easy verification in place making the process more efficient & global.

d. **Speed:** Automated verification and issuance processes take days, not months as compared with manual systems.

Disadvantages

a. **Centralized Control:** Vulnerable to cyberattacks, dependency on a centralized database leads to it being the weakest link which could be breached and exposed.

b. **Data Breach Risks:** If the centralized database is hacked, it can undermine certificate authenticity and privacy.

c. **Digital Divide:** Unless all the stakeholders are well-equipped, both in terms of digital literacy and technology access to use these systems, efficacy of these poles can be blunt.

d. **Interoperability Issues:** Since different organizations use different standards and technology for their digital certificates, there are interoperability issues, which refrain them from verifying the user identity in a seamless manner.

Both manual paper-based systems and non-blockchain digital certificate systems have their own set of pros and cons. Manual systems, while old hat and simple in the digital age, typically face security threats or logistics problems. At the same time, digital non-blockchain solutions are less centralized, less risky in comparison with centralized systems and being more reliable and convenient to use, they contain risks involved in centralized data storage on one hand, and accessibility bottlenecks on the other. These existing systems are the bottom line in which researchers can further explore solutions to how blockchain technology may overcome these limitations and improve certificate issuance, and authentication in Nigerian universities. With the decentralized and hash-protected technology of blockchain, universities can dramatically increase efficiency in certification security and urgency at scale.

2.2 BLOCKCHAIN TECHNOLOGY

A blockchain is a distributed ledger that records all transactions or digital events shared among a network of participants. Once information is entered into this database, it becomes permanent and unchangeable. Blockchain technology facilitates trust and transactions within large peer-to-peer networks without the need for centralized authority. Each digital transaction is grouped into a block, and these blocks are linked together to form a continuous chain, known as the blockchain. The contents of these blocks can be either pre-agreed upon or randomly generated by users, but once a block is added through consensus, it cannot be modified, ensuring the immutability of the data (Ali, 2022). Blockchain offers several key benefits, including provenance, consensus, and finality, providing all participants with an up-to-date ledger (Effiong, 2020). To maintain confidentiality and integrity, blockchain technology employs public cryptographic mechanisms, utilizing both public and private keys for encryption and decryption. These cryptographic keys also allow the generation of digital signatures, which help verify the integrity of the data (Antonopoulos, 2021). The foundation of blockchain technology lies in cryptography and distributed computing, which together create a secure and decentralized system for recording and verifying transactions. By integrating cryptography for data security and distributed computing for consensus mechanisms, blockchain has transformed the way information is stored and shared in environments where trust is a concern (Benet, 2019). This innovative technology has opened the door to new applications across a wide range of industries, from finance to supply chain management, with the potential to redefine traditional processes and significantly enhance efficiency (Techopedia, 2024).

2.2.1 TYPES OF BLOCKCHAIN

I. Public Blockchain

Public blockchains are accessible to everyone and do not require permissions to participate. These blockchains can employ different consensus mechanisms, but scalability can become a challenge as the user base expands. Notable examples of public blockchains include Bitcoin and Ethereum, with other well-known networks like Litecoin, Ripple, and Dogecoin also falling under this category (Antonopoulos, 2021). Despite the scalability concerns, public blockchains offer significant advantages such as transparency, decentralization, and security, making them a favored choice for users who prioritize these attributes in their transactions.

II. Private Blockchain

Private blockchains cater to the needs of enterprises, providing a permissioned environment limited to select individuals, typically within business entities. Hyperledger Fabric stands out as a prominent example of such private blockchains. They offer heightened security and data control, especially beneficial for sectors like finance, healthcare, and supply chain management (Mendling et al., 2021). These private networks enable the secure sharing of sensitive information, ensuring that access is restricted to authorized entities only. Businesses frequently opt for Hyperledger Fabric due to its modular design and support for smart contracts, which allow them to harness the benefits of blockchain technology in a controlled and efficient environment (Bach et al., 2020).

III. Hybrid Blockchains

Public and private blockchains each come with their own set of advantages and disadvantages. The concept of a hybrid blockchain, exemplified by platforms like

Dragonchain, seeks to merge the strengths of both types. Hybrid blockchains offer the security and transparency associated with public blockchains while providing the control and privacy features of private blockchains. This combination makes them adaptable for a wide range of industries and use cases (Koens & Poll, 2019). As Dragonchain leads the way in this domain, it is expected that more organizations will adopt hybrid blockchains in the near future.

IV. Consortium Blockchains

Consortium blockchains, such as R3 Corda, are jointly owned and managed by multiple private nodes within a partnership. They serve as a middle ground between public and private blockchains, enabling trusted entities to collaborate while retaining a degree of control over network operations. Compared to fully public blockchains, consortium blockchains offer enhanced privacy, scalability, and efficiency, making them particularly well-suited for industries such as finance, supply chain management, and healthcare. The collaborative governance model of consortium blockchains like R3 Corda promotes transparency, security, and interoperability among participants, which helps streamline business operations and opens up new opportunities for innovation and value creation.

2.2.2 BLOCKCHAIN ARCHITECTURE

Blockchain architecture consists of several components and subcomponents, or layers, that together form a complete and functional blockchain system. Each layer has a distinct role, ranging from data storage to network maintenance and ensuring consensus across the system. Understanding these layers is crucial for blockchain developers and start-up entrepreneurs who want to effectively utilize blockchain technology.

1. **Ledger Structure:**

A blockchain is essentially a ledger that links sequential "blocks" of transactions. Here's how the process works:

i. **Network Access:** Anyone who wants to trade assets across a private or public network needs access to that network. This access is provided through a software application known as a "wallet," which acts as an intermediary between the user and the blockchain. A wallet can be installed on a device or accessed via a web browser. Depending on its design, a blockchain wallet can send and/or receive digital assets. Some wallets enable direct transactions without the need for a third party, while others are managed by third parties responsible for safeguarding users' digital assets.

ii. **Transaction Validation and Consensus:** Users interested in validating transactions typically need to install blockchain software on their devices. This software is responsible for writing transactions to the ledger, maintaining a complete copy of the ledger, and ensuring that all copies stay synchronized across the network. In public blockchains, anyone can install the software and keep a full ledger copy, which allows direct participation in the blockchain network without any third-party conditions. In permissioned blockchains, however, a central authority decides who can run a node and take part in the consensus process.

iii. **Cryptographic Security:** Transactions (blocks) in a blockchain are cryptographically linked, making them tamper-proof. Once a transaction is recorded and time-stamped on the blockchain, it becomes immutable, unlike records in traditional digital databases that can be altered or deleted.

2. Transaction Documentation:

iv. **Transaction Details:** The blockchain documents every transaction's details, including what was transferred, the parties involved, any structured information (metadata) about the transaction, and a cryptographic hash or "digital fingerprint" of the transaction content. This unique signature ensures the transaction's verification: if any part of the transaction content is altered, its unique hash will no longer match the one stored on the blockchain, alerting the system to the discrepancy.

3. **Consensus Mechanisms:**

v. **Consensus Process:** Before a new transaction record is added to the blockchain, all involved parties must reach a consensus. The network nodes validate that these parties have the necessary authorization to perform the transaction. Once one party agrees to send an asset, another agrees to receive it, and the nodes confirm their transaction capabilities, the transaction is finalized.

4. **Network Integrity and Security:**

vi. **Continuous Validation and Security:** All computers (nodes) in the blockchain network continuously validate the integrity of their blockchain copy against all other copies in the network through mathematical algorithms. The version of the blockchain that is prevalent on most computers is considered the authentic one, making it nearly impossible to "hack" the records unless one controls over half of the network's computers. With public blockchains like Bitcoin and Ethereum running on thousands, or potentially millions, of computers, it would require an enormous effort to destroy the ledger entirely, as it would necessitate deleting every existing copy worldwide.

This robust architecture ensures that blockchain systems are secure, transparent, and resilient, making them suitable for a wide range of applications across different industries.

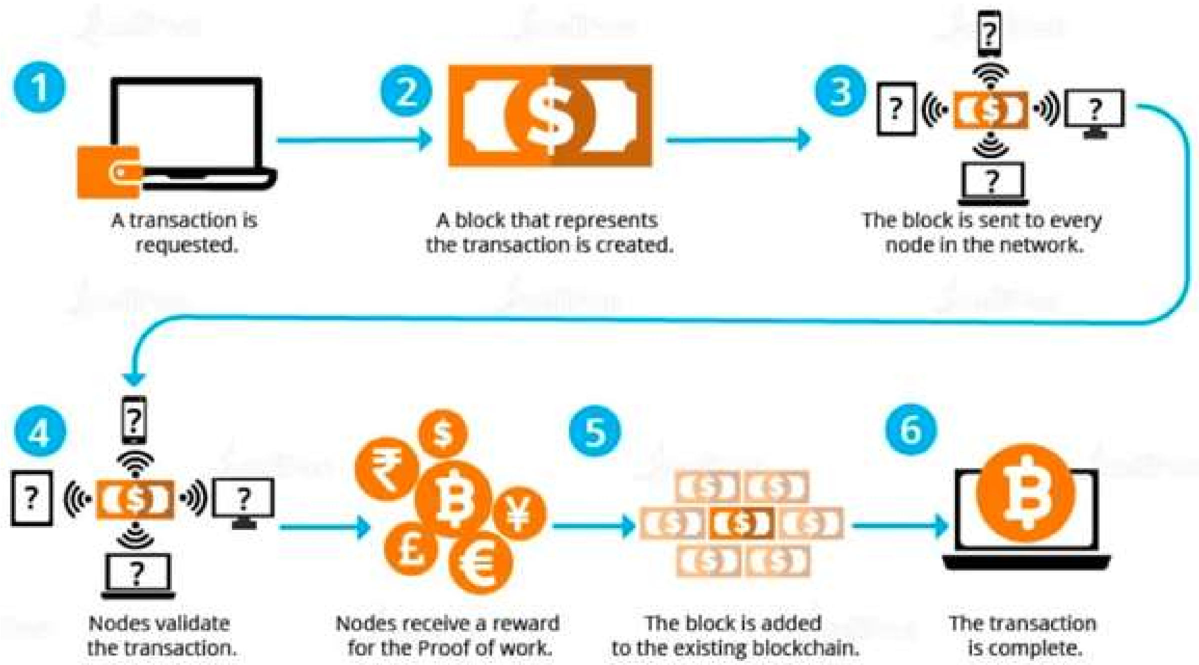

Figure 2.0.1 ***Active structure of blockchain (Ali, 2022)***

2.2.3 APPLICATIONS OF BLOCKCHAIN TECHNOLOGY

2.1.3.1 Blockchain for Digital Identity

Blockchain-based identity verification could become a key method for maintaining the integrity of user identities. This approach would also allow companies to avoid the costs associated with storing user data in traditional databases. Additionally, users benefit from the convenience of easily accessing and having greater control over their identity data. Blockchain technology enhances security by leveraging its decentralized structure, making it more challenging for hackers to breach sensitive identity information. By utilizing blockchain-based identity verification, the risk of identity theft and fraud can be significantly minimized, offering peace of mind to both companies and users. The potential applications of blockchain in identity verification are extensive, presenting a promising solution to the challenges of managing and protecting identity in the digital age (Wang, 2019).

2.1.3.2 Blockchain for Cryptocurrency

Bitcoin is the initial cryptocurrency that introduced distributed ledger technology and was created in 2009. Since then, it has garnered popularity among business owners seeking a decentralized trust model for their transactions. As the adoption of digital currencies grows, Bitcoin continues to stand out as a trailblazer in the crypto space. Its groundbreaking blockchain technology has paved the way for the development of countless other cryptocurrencies, each offering distinct features and capabilities. Despite encountering challenges and experiencing fluctuations in value, Bitcoin remains a powerful and influential presence in both the financial and technological realms.

2.1.3.3 Blockchain Technology for Certificate Verification

Verification of documents is important when handling counterfeit certificates. In a shared environment, safeguarding the authenticity of certificates is crucial to prevent tampering and the misuse of fake certificates as legitimate ones, resulting in faster and more efficient verification processes. This is particularly vital in industries where certification is mandatory, such as healthcare and finance, where the undetected use of counterfeit certificates could have serious repercussions (Effiong, 2020). By implementing robust verification procedures and leveraging advanced technology, organizations can preserve the integrity of their certificates and maintain trust in the certification system (Omijeh, 2023). Regular training and awareness programs for staff members can also help in identifying potential fake certificates and taking necessary actions to address the issue promptly. Ultimately, verification of documents is a fundamental step in safeguarding the credibility and reliability of certificates in any given setting.

2.1.3.4 Blockchain for Real Estate

The real estate industry has many players and a large degree of intermediation; thus, trust is the main element that defines the speed of transactions. One way of creating confidence is following the involvement of blockchain technology in maintaining records of properties in a reversible manner. It would be helpful to devise a register that contains all the details of the property from different owners and other important references to avoid any delay so that every record is easily accessible whenever the need arises. By so doing it is possible to greatly lower fraudulent actions and property transactions processes. In addition, the technology allows the use of smart contracts that can facilitate and enforce conditions of agreements between the buyer, the seller, and probably several other parties involved. This automation can in turn shorten the amount of time it takes to do a transaction, cut down on the paperwork as well as increase the level of security in the transactions in real estate business. In sum, the use of blockchain technology will bring tremendous changes to the form and function of Real Estate immeasurably by building the trust amongst all the parties involved in the process as well as improving the efficiency and security measures in this industry (Reyes, 2021).

2.2.4 BLOCKCHAIN PLATFORMS

There are various methods for creating a blockchain, with the most significant ones being:

I. Bitcoin

Bitcoin operates as a digital payment system that relies on cryptographic proof instead of trust, allowing parties to conduct transactions directly without the need for a trusted intermediary. Created by the pseudonymous Satoshi Nakamoto, Bitcoin represents the first application of blockchain technology. Today, the Bitcoin network is the most extensive public blockchain in existence (Nakamoto, 2019). Bitcoin acts as a digital equivalent to physical cash. While physical cash is authenticated through its tangible characteristics like serial numbers and security features on banknotes, it lacks a ledger to document transactions, leading to issues like counterfeiting. In contrast, Bitcoins rely on a transaction ledger for verification. Both physical cash and Bitcoins demand secure storage, be it in physical or virtual wallets, as negligence in safeguarding them can result in theft (Antonopoulos, 2021).

The Bitcoin blockchain allows embedding up to 80-character strings in each transaction, making it ideal for storing document hashes on a public record. This feature upholds the integrity of digital signatures, rendering them tamper-proof. Being an open-source endeavour, Bitcoin is overseen by its user community. Modifications to the Bitcoin software, protocol, and blockchain are endorsed once a majority of the network's computers transition to the updated software version. Nonetheless, a few downsides of the Bitcoin blockchain include:

i. Recording solely the sender, recipient, transferred amount, and a hash.
ii. Handling less than 10 transactions per second, a fraction of what standard credit card networks can manage.
iii. Experiencing exponential growth in size, making it difficult for users lacking significant computing power to uphold a complete blockchain copy. This issue reduces network involvement and compromises overall.

II. Ethereum

Ethereum, released in 2015, is a block-chain platform and distributed computing system. It is known for its smart contracts written in Solidity. These contracts are compiled into bytecode that operates on the Ethereum Virtual Machine (EVM) (Buterin, 2020).

Components of Ethereum

a. Smart Contracts

Smart contracts are concise computer programs embedded in a blockchain that automatically execute transactions when specific conditions are met. In essence, a smart contract functions like a conditional statement, such as "transfer X to Y if Z occurs." Unlike traditional contracts, which require manual execution after an agreement is made, smart contracts are self-executing. Once programmed into a blockchain, the transaction is automatically carried out as soon as the predefined conditions are fulfilled. The primary goal of smart contracts is to reduce the need for trusted intermediaries, thereby lowering arbitration and enforcement costs, minimizing fraud-related losses, and mitigating both intentional and unintentional errors. The Ethereum network is a notable example of a blockchain that heavily relies on smart contracts. It features a virtual machine that executes code across all network nodes and charges fees for these services (Buterin, 2020).

A pivotal element of the Ethereum network is the Ethereum Virtual Machine (EVM). The EVM serves as the operational environment for smart contracts within Ethereum. It operates in a sandboxed and fully isolated manner, ensuring that code within the EVM remains cut off from the network, file system, and other processes. Smart contracts also possess restricted access to other smart contracts, guaranteeing a high degree of security and trustworthiness in the actions they undertake (Wood, 2019).

b. Solidity

Solidity is the programming language commonly used for smart contracts on Ethereum and other blockchains. Influenced by C++, Python, and JavaScript, Solidity targets the EVM. It is an object-oriented, statically typed high-level language. Solidity supports complex variables like hierarchical mappings and structs for contracts, along with inheritance including multiple inheritances following C3 linearization. It uses an ECMAScript-like syntax, familiar to web developers, with static typing and variadic return types. Solidity introduces an ABI for multiple type-safe functions in a contract, enabling the creation of advanced contracts for various purposes such as voting, crowdfunding, blind auctions, multi-signature wallets, and more (Dannen, 2019).

c. Application Binary Interface (ABI)

The application binary interface (ABI) in computing functions as a crucial link between two binary program modules, facilitating their interaction and ensuring compatibility. Typically, one module is a library or operating system feature, while the other is a program executed by a user. The Application Binary Interface (ABI) outlines the procedures and conventions by which data structures and computational routines are accessed and managed at the machine code level, which is inherently tied to the underlying hardware. Conversely, the Application Programming Interface (API) defines the interaction with these elements within the source code, offering a higher-level, hardware-agnostic, and typically human-readable framework. A key aspect of an ABI is the calling convention, which establishes the protocol for passing data as arguments to computational routines and retrieving results. For instance, the x86 calling conventions regulate the manner in which functions handle parameters and return values at the machine code level. Compliance with an ABI, whether it is formally standardized or not, is generally the responsibility of the compiler, operating system, or library developers. However, application programmers may need to engage directly with an ABI when working on programs that involve multiple programming languages or when compiling a single language program using different compilers (Ethdocs, 2020).

III. Hyperledger Fabric

Hyperledger Fabric, originally launched under the Linux Foundation and now primarily driven by IBM and Digital Asset, is designed to deliver a robust, adaptable, and secure blockchain framework. It is widely recognized as a leading platform for developing private, open-source blockchains tailored to business environments (Androulaki et al., 2019).

Despite its strengths, Fabric's architecture is notably more intricate than other blockchain platforms, potentially reducing its resilience against tampering and attacks. Although it is categorized as a "private" blockchain, Fabric faces challenges in scalability and performance. Consequently, projects based on Fabric may encounter complex and potentially vulnerable deployments that struggle to meet the operational requirements effectively (Mendling et al., 2021).

2.2.5 IDES

Several Smart Contract IDEs exist, the most popular among them are:

a. Remix: A web browser-based IDE to write Solidity smart contracts. This is where you can write, deploy, and run them. The most straightforward approach to experimenting with Solidity is through Remix. This platform provides a user-friendly interface equipped with features such as auto-completion, syntax highlighting, and error checking, which streamline the process of writing Solidity contracts (Consensys, 2020). Moreover, Remix enables the testing of Smart Contracts within a simulated environment before actual deployment on the blockchain, ensuring they operate correctly and efficiently.

b. Ganache: Ganache is a smart contract IDE providing a personal blockchain for Ethereum DApp development. It supports Windows, Mac, and Linux, allowing safe testing and deployment with control over block times, gas prices, and blockchain events. As part of Truffle Suite, it helps developers of all levels streamline their development process (Truffle Suite, 2020).

c. Truffle: Truffle is a development environment, testing framework, and asset pipeline for Ethereum blockchains using the EVM bytecode. It makes blockchain development easier by offering tools, libraries, and integrations for creating smart contracts and DApps that can be published and developed on the Ethereum platform.

2.2.6 IPFS

The Interplanetary File System (IPFS) is an open-source, decentralized network designed to connect all computers within a unified file system. By leveraging peer-to-peer technology, IPFS allows users to share and store files across a distributed network, enabling greater efficiency, resilience, and accessibility compared to traditional centralized systems. When a file gets stored inside IPFS, it will get a unique hash address to fetch that file. It provides file versioning and deduplication all in one. It is a decentralized web version that comes with a unique concept to push static files and, as a result, cut down the costs of applications and improve response times (Benet, 2019).

2.2.7 POLYGON

Polygon, often known as Matic, is a cryptocurrency token symbolized by MATIC and represents a technology platform developed to deliver scaling solutions for the Ethereum Network and other blockchain ecosystems. Originally launched as Matic Network in 2017, Polygon has evolved into a comprehensive full-stack scaling solution designed to enhance the performance and scalability of blockchain networks. The platform provides various tools and protocols that significantly increase transaction speeds and lower costs, establishing it as a vital element within the broader Ethereum ecosystem. It was created as an integration on the Ethereum blockchain, designed to enhance the ecosystem that is Ethereum-compatible. Polygon, is an open-source platform for the development and integration of blockchain networks, is able to provide virtually infinite possibilities for functionality and autonomy of any blockchain project while also ensuring the security, environmental friendliness, and mass compatibility of the blockchain with a platform that has been tested by millions of users—Ethereum (Polygon, 2021).

2.3 REVIEW OF RELATED WORKS

A thorough understanding of related studies is crucial for research, as it forms the foundation for future exploration. This understanding enables researchers to grasp what has already been achieved in the field and to pinpoint the knowledge gaps that their current research seeks to fill. By building on prior work, researchers can ensure that their studies are relevant, well-informed, and contribute significantly to the existing body of knowledge. This section present background studies and to explore further the reviews that are directly related to the objectives of this paper.

Blockchain-Based Document Verification System by Ali Falah Hassan and Ali Hussein Salim (2022)

A significant conclusion from the literature is that blockchain technology enables the realization of strict, scalable, and secure solutions in academic certificate verification processes. It further distributes the process even more through decentralization, thus eliminating third parties in the verification process to cut costs and time. The system automates certificate creation and verification via Binance Smart Chain. Meanwhile, the hashing of certificates is made, a QR code is provided for verification, and in case of some changes, hashed values of certificates are saved in the blockchain file system. It reduces dramatically the risk of loss of the certificate, as well as the integrity of the data, since any change of the certificate results in a different hash value. First, the project proves that the data is safe due to the application of a practical SHA-3 hash function, by which it is almost impossible for one to regenerate the original document out of the hash.

However, despite the system's benefits, it has shortcomings. The first significant challenge is that the system still relies on the Binance Smart Chain, which in itself turns out to be cost-effective when compared to Ethereum. Yet, it also suffers from network congestion and various transaction costs that can be very high in high usage. Moreover, dependence on BSC suggests that the system may never entirely exploit all the interconnectivity and broader support of ecosystems available from other blockchain networks.

To overcome these drawbacks, the use of the Polygon network is proposed. Polygon is a scalable alternative that is cost-effective, reduces the fees for transactions, and guarantees efficiency in the system. This system remains highly efficient, even during peak hours, with lower-cost transactions, thanks to the reliance on these Polygon Layer 2 solutions. Another thing is the compatibility of Polygon with the Ethereum ecosystem: once it is exposed, there will be greater interoperability and access to a more extensive range of decentralized applications and services, augmenting the functionality and the experience of using the system.

This base paper serves as the foundation of my research, establishing the core principles and framework upon which we propose enhancements through the integration of the Polygon network. This project work builds upon these findings to address the identified challenges and further advance the academic certificate verification process using blockchain technology.

Blockcerts Project by Massachusetts Institute of Technology Media Lab, led by J. Philipp Schmidt and Juliana Nazare (2021)

MIT (Massachusetts Institute of Technology) has introduced a new standard for issuing and confirming certificates using the Bitcoin blockchain. This innovative system leverages blockchain technology to create a decentralized and tamper-proof method for managing educational credentials. The primary application for this new standard is known as the Blockcerts Wallet, an app that allows students to store, access, and share their diplomas securely.

The Blockcerts Wallet enables students to obtain a trustworthy and unalterable version of their degree, which can be shared with various stakeholders such as employers, academic institutions, family, and friends. This advancement marks a major step forward in the field of digital certificates, enhancing the authenticity and integrity of academic credentials. The process starts with the issuer, typically an educational institution, digitally signing the certificate and recording it, along with its hash values, on the blockchain. This certificate is then securely assigned to the intended recipient, ensuring that both the ownership and issuance of the credential are transparent and easily verifiable.

Though, the implementation of blockchain technology for educational credentials is not without challenges. One significant issue with the Blockcert Wallet is the aspect of ownership. Certificates are generated based on the key pairs created by the users, which means that the management and security of these keys are important. If a user loses their private key, they lose access to their credentials, which poses a risk that needs to be managed effectively. Privacy is another critical consideration in this system. While blockchain's transparency allows everyone to access the data, the certificates are designed to be useful only to the specific user they are issued to. This creates a potential conflict, as the open nature of blockchain can expose sensitive information if not handled correctly. Blockcerts is a nice system but it does not provide for the authenticity of issuing institutions. This turns out to be a major problem that anyone in the network can issue certificates.

A Framework for the Adoption of Blockchain Technology in Academic Certificate-Verification Systems: A Case Study of Nigeria; Mercy Effiong (2020)

Effiong's thesis, titled "A Framework for the Adoption of Blockchain Technology in Academic Certificate-Verification Systems: A Case Study of Nigeria," explores the considerable challenges involved in verifying academic certificates in Nigeria. The research underscores the inefficiencies of the existing paper-based verification systems, which are cumbersome, unreliable, and time-consuming. Effiong contends that these manual processes result in elevated administrative costs, delayed responses, and widespread dissatisfaction among stakeholders.

The study reveals the widespread prevalence of fraudulent academic credentials and the inadequacy of traditional verification methods in combating this issue. Relying on paper-based databases and documentation renders the verification process inefficient and susceptible to errors. Effiong stresses that leveraging blockchain technology can provide a more efficient, transparent, and secure solution. The decentralized nature of blockchain and its capacity to offer verifiable, tamper-proof records present a promising alternative to current practices.

Effiong identifies several challenges that could hinder the adoption of blockchain technology in Nigeria, including unreliable electricity supply, inadequate ICT infrastructure, funding constraints, a lack of digital skills, and limited awareness of blockchain technology. Additionally, resistance to change stemming from fears of displacement, technophobia, and corruption poses a significant barrier. However, many of these challenges have diminished in relevance due to the Central Bank of Nigeria's adoption of blockchain technology with the launch of the eNaira. Currently, it is estimated that over 22 million people, or 10.3% of Nigeria’s total population, own cryptocurrency

(Triple A, 2022). KuCoin, a prominent global cryptocurrency exchange, has revealed that Nigerians are among the most avid cryptocurrency adopters worldwide. For many Nigerians, cryptocurrency has become the preferred method for storing and transferring assets. Remarkably, Nigeria experienced a 2,467.2% surge in cryptocurrency users in 2021 alone (KuCoin, 2022).

Effiong contends that despite the significant barriers to implementation, the benefits of a blockchain-based verification system namely cost reduction, increased efficiency, and enhanced security make it a compelling solution for improving academic certificate verification in Nigeria. These advantages outweigh the challenges, positioning blockchain as a promising technology to modernize and streamline the verification process within the educational sector. The research offers an in-depth examination of the deficiencies in Nigeria's current academic certificate verification system and proposes a robust framework for integrating blockchain technology to tackle these challenges. It underscores blockchain's potential to transform academic credential verification, enhancing reliability and efficiency, while also acknowledging the hurdles that need to be overcome for successful implementation.

Development And Implementation Of Blockchain Solution To Enable Data Security At University Of Port Harcourt; Enoch, J. D., Omijeh, B. O., & Okeke, R. O. (2023)

The study proposed a blockchain-based solution aimed at improving the security and efficiency of managing academic credentials at the University of Port Harcourt. The proposed system includes several key components. The blockchain data structure was originally designed to securely store academic certificates, incorporating key features such as proof of work, block mining, transaction creation, chain validation, and data retrieval. These elements collectively ensure the security and integrity of the stored information.

To further strengthen data security, a decentralized blockchain network was established, consisting of multiple API servers that interact and share data to maintain synchronization. This network architecture aims to eliminate single points of failure, thereby enhancing the robustness and security of the entire system. A consensus algorithm was implemented to keep the network nodes synchronized, ensuring that all nodes maintain consistent and valid data, which in turn bolsters the overall integrity of the blockchain. Additionally, the Blockchain Explorer application was developed to provide a user-friendly interface for querying the blockchain. This tool enables university authorities, employers, and other stakeholders to efficiently and securely verify specific blocks, results, transcripts, or certificates.

However, the development and deployment of a decentralized blockchain network with multiple API servers proved to be both complex and resource-intensive. The technical expertise required to establish and maintain such a system posed significant challenges for broader adoption across other institutions. As the volume of transactions and blocks increases, the system may face scalability challenges. Continuous monitoring and optimization are essential to ensure the network remains efficient and responsive under heavy load. The study proposes a solution only applicable to the university of Port Harcourt and will be difficult to be implemented in other universities in Nigeria. This is because each university in Nigeria has its unique challenges and variations in systems and structures. Therefore, what may work effectively at the University of Port Harcourt may not necessarily be suitable for other universities in the country. It is important for a central body like the NUC to customize solutions that will be universally effective for all universities in Nigeria.

Development of blockchain-based computerized transcript processing system; Ezema, L. S., Ezeh, G. N., Onwuatuegwu, H. U., & Bede-Ugwu, J. P. (2024).

The study introduces a computerized transcript processing system that utilizes a multi-tiered architecture to deliver a streamlined and efficient user experience. This architecture consists of several distinct layers, each playing a crucial role in the system's overall functionality and security.

The **Frontend Layer (Presentation)** provides an intuitive user interface, enabling easy interaction for both students and administrators. This layer is designed to facilitate tasks such as requesting transcripts, accessing various system features, and navigating the platform seamlessly, ensuring that the user experience is both accessible and efficient.

The **Backend Layer (Application)** is responsible for managing business logic and data processing. It acts as the intermediary between the frontend and other components of the system, ensuring that data flows efficiently and that requests from the frontend are handled appropriately. This layer is essential for maintaining the system's functionality and overall performance, as it processes core operations and ensures the seamless execution of tasks.

The **Database Layer** handles the storage and management of critical data, including student information, course details, and transcripts. As the central data repository, this layer ensures that all records are securely stored and can be easily retrieved when needed. The system uses MongoDB, a distributed NoSQL database, which facilitates efficient data storage and retrieval through its BSON format and unique Object IDs. However, while MongoDB offers speed and flexibility, it also raises some security concerns, such as vulnerabilities to SQL injection attacks, which require careful management.

The **Smart Contract Layer**, deployed on the Binance blockchain, manages the secure processing of transcript fees. By utilizing blockchain technology, this layer ensures that financial transactions are transparent and secure. Smart contracts automate fee transactions and maintain a tamper-proof record of all payments, thus enhancing the integrity and security of the payment process.

Despite its innovative approach, the system faces several challenges and limitations. One of the primary challenges is the **complexity of implementation**. Developing and deploying a decentralized blockchain network with multiple API servers is resource-intensive and requires significant technical expertise, potentially hindering broader adoption across other institutions.

Scalability is another issue, as the system may struggle to maintain efficiency and responsiveness as the number of transactions and blocks increases. Continuous monitoring and optimization are necessary to ensure the network performs well under heavy load, which can be resource-demanding.

The reliance on MongoDB for data storage, while efficient, introduces potential **security vulnerabilities**, such as exposure to SQL injection attacks. Unlike decentralized storage solutions like the InterPlanetary File System (IPFS), MongoDB does not inherently offer the same level of data immutability and decentralized security, potentially risking data integrity.

Lastly, **customization** presents a significant challenge. The proposed solution is specifically tailored for the Federal University of Technology, Owerri, and adapting it for use in other universities may prove difficult due to varying institutional needs and existing system structures. A more universal solution, possibly coordinated by a central body like the National Universities Commission (NUC), might be necessary to create an effective system that can be adopted across all Nigerian universities.

.

CHAPTER THREE

METHODOLOGY

3.1 INTRODUCTION

Now that we have a basic understanding of blockchain technology and the concepts behind it, how it is developed, and the practical tools required to implement it from the previous section, we can finally begin to conceptualize and design the platform we want to build. In this chapter, we will explore the techniques used in creating a certificate validation system. This chapter focuses on the processes and tools used during development.

3.2 RESEARCH DESIGN

A blockchain-based platform for verification and authentication is being designed as the research project. The design effectively focuses on creating a secure and useful application to combat certificate forgeries and authentication issues in Nigerian universities. This is a very practical application of blockchain technology made possible by the software development approach, providing a solution to design and program a tool that offers the desired security and usability.

3.3 REQUIREMENT ANALYSIS

3.3.1 FUNCTIONAL REQUIREMENTS

This section discusses the general platform’s functional requirement and the platforms smart contract requirements.

These are the specific behaviours of the platform. These include:

I. **Metamask Wallet Integration:**

a. NUC and university administrators will have to sign in with their Metamask wallet to log in to the platform.

b. Securely integrate and connect to the blockchain with Metamask.

II. Academic Certificates Issuance:

a. Universities must provide digital issuance of academic certificates through the platform.

b. Every certificate should be recorded on the blockchain with a unique digital signature and hash.

III. Certificate Storage:

a. All academic certificates will be stored on IPFS to ensure decentralized and secure storage.

b. The system must produce a different hash for each on-chain certificate and store it on IPFS.

IV. Certificate Verification:

a. This function will allow employers, schools, etc., to verify the genuineness of the academic certificates on the platform.

b. Verification is expected to be real-time and should use the hashes stored on the blockchain to fetch and validate certificates located on IPFS.

V. Implementation of Smart Contracts:

a. Smart contracts are needed to standardize the processes of issuing and verifying certificates, ensuring tamper-proof transactions.

b. The system should enable the development, deployment, and management of smart contracts for different academic certificate-associated operations.

VI. User Interface:

a. Provide a welcoming and easy-to-use interface for university administrators and verifiers.

b. The interface should support certificate requests, issuance, verification, and status tracking.

VII. Data Privacy and Security:

a. Ensure academic certificates are encrypted and stored securely.

b. Enforce protections against unauthorized access and data breaches.

3.3.2 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements represent the overall performance and usability of the system in specific functionalities.

I. Performance:

a. The system should handle a high number of transactions simultaneously without lag.

b. Enable fast IPFS/blockchain certificate lookup and verification.

II. Scalability:

a. The system should scale to support increasing numbers of users and certificates as more universities join.

b. Support horizontal and vertical scaling to handle large increases in load while maintaining performance.

III. Reliability:

a. Ensure high platform availability with minimal downtime.

b. Implement failover mechanisms and backups to prevent data loss.

IV. Usability:

a. The platform should be clear and intuitive to use.

b. Provide detailed user manuals and support documentation.

V. Security:

a. Follow security best practices, deploy encryption, and interact with the blockchain safely.

b. Update and patch the system periodically to fix security flaws.

3.4 SYSTEM MODELING

3.4.1 WORKFLOW DIAGRAM

Workflow diagrams help to map out a series of steps or tasks to complete a process. They are frequently used to depict the sequence of stages in a project or in a business process. These diagrams assist in recognizing inefficiencies, repetitions, and areas for improvement within the workflow. Finally, and of greater importance, they serve as a great way to share elaborate processes with other team members or stakeholders.

The first thing to start off is to build a smart contract, which is the genesis of building the blockchain system. The smart contract will be used to build the front-end. After that, we link the smart contract and the front-end with Web3. Transactions are done with digital wallets (such as Metamask, which is a type of digital wallet application). This is described in the workflow diagram below:

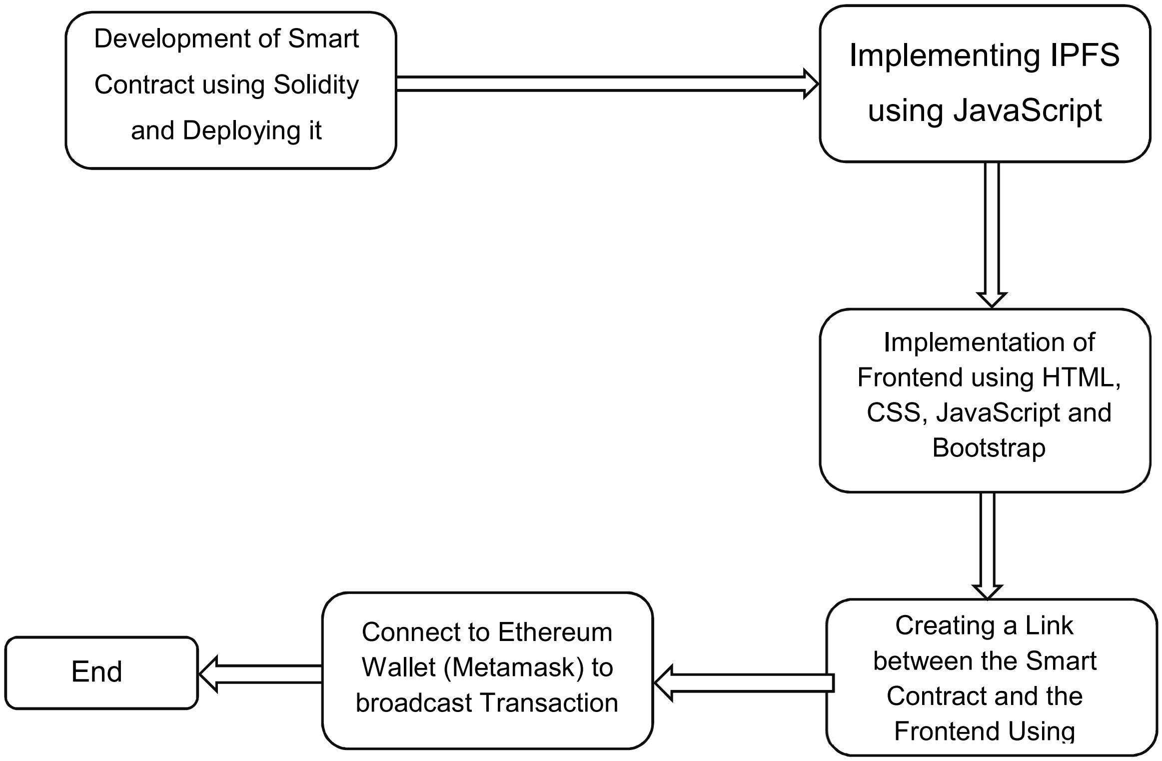

Figure 3.0.1 Workflow of the implementation

3.4.2 FLOW CHART

A flowchart is a visualization of a particular process, showing the steps involved and the order in which they are implemented. It uses standardized symbols to represent the respective actions, decisions, and outcomes. Flowcharts are widely used in business and engineering to document, order, and clarify processes, workflows, and algorithms. They are a way to visualize, document, and analyse processes to easily spot inefficiencies, bottlenecks, and improvement spots.

The core of the Ethereum blockchain is the smart contract. Remix is used for creating Solidity smart contracts and deploying them. Currently, Remix is the best Ethereum IDE for experimenting with Solidity smart contracts.

The smart contract serves multiple functions:

1. **Addition and modification of Certificate exporters (Universities):** The owner of the smart contract (NUC) can add and modify Certificate exporters (Universities) representing the university's certificate issuing administrator. This function is called when NUC (contract owner) adds or registers a university to the blockchain. This is demonstrated in the flowchart below:

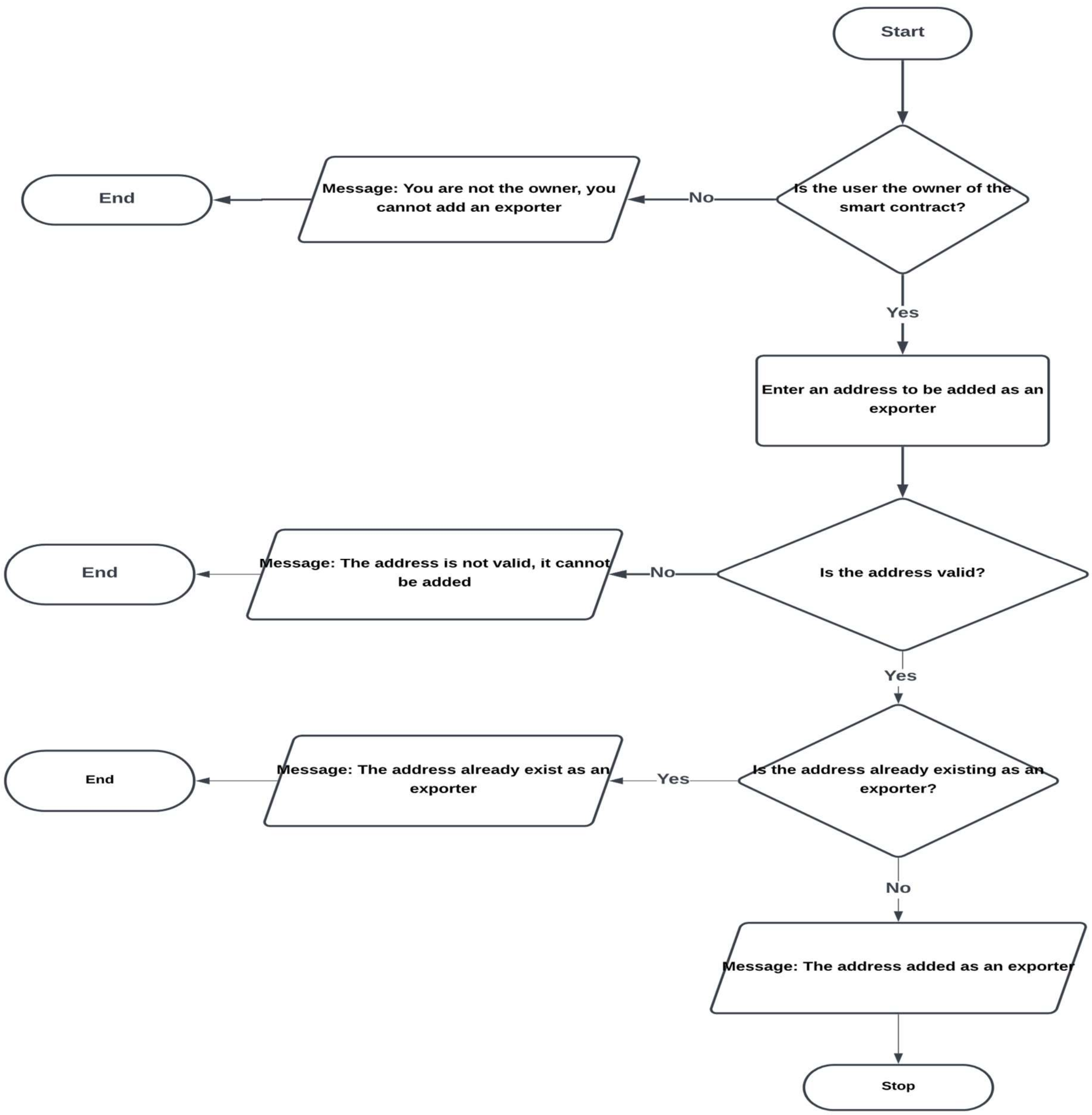

Figure 3.0.2 Flowchart of Adding Certificate Exporters

2. **Deletion of a certificate exporter (university): The** smart contract owner (NUC) can delete a certificate exporter (university). If the owner deletes the exporter, they will not be able to upload documents to the blockchain. This is to address cases of compromised university wallets. This process is depicted below:

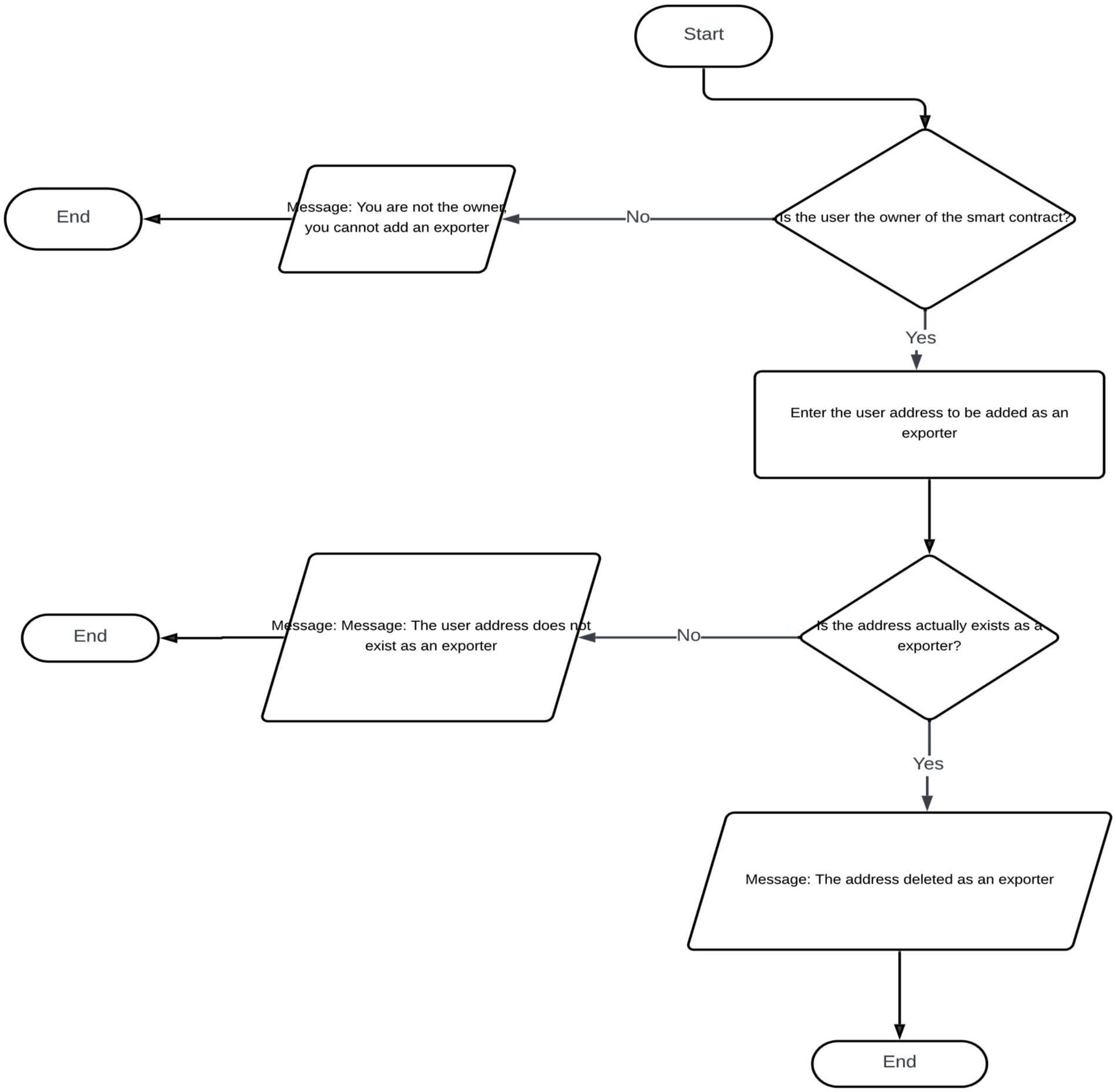

Figure 3.0.3 Process of Deleting a Certificate Exporter

3. **Certificate upload:** This function is exclusive to universities (Exporters) for uploading a university student's certificates to the blockchain and providing a QR code to the student. This is demonstrated in the flow chart below.
4. **Document verification**: This function verifies the authenticity of documents issued by a specific university when a verifier requests verification of a certificate. The flowchart is depicted below
5. **Certificate Revoking/Deletion:** The owner or the issuing university has the authority to delete a previously uploaded certificate. The flowchart illustration is shown below.

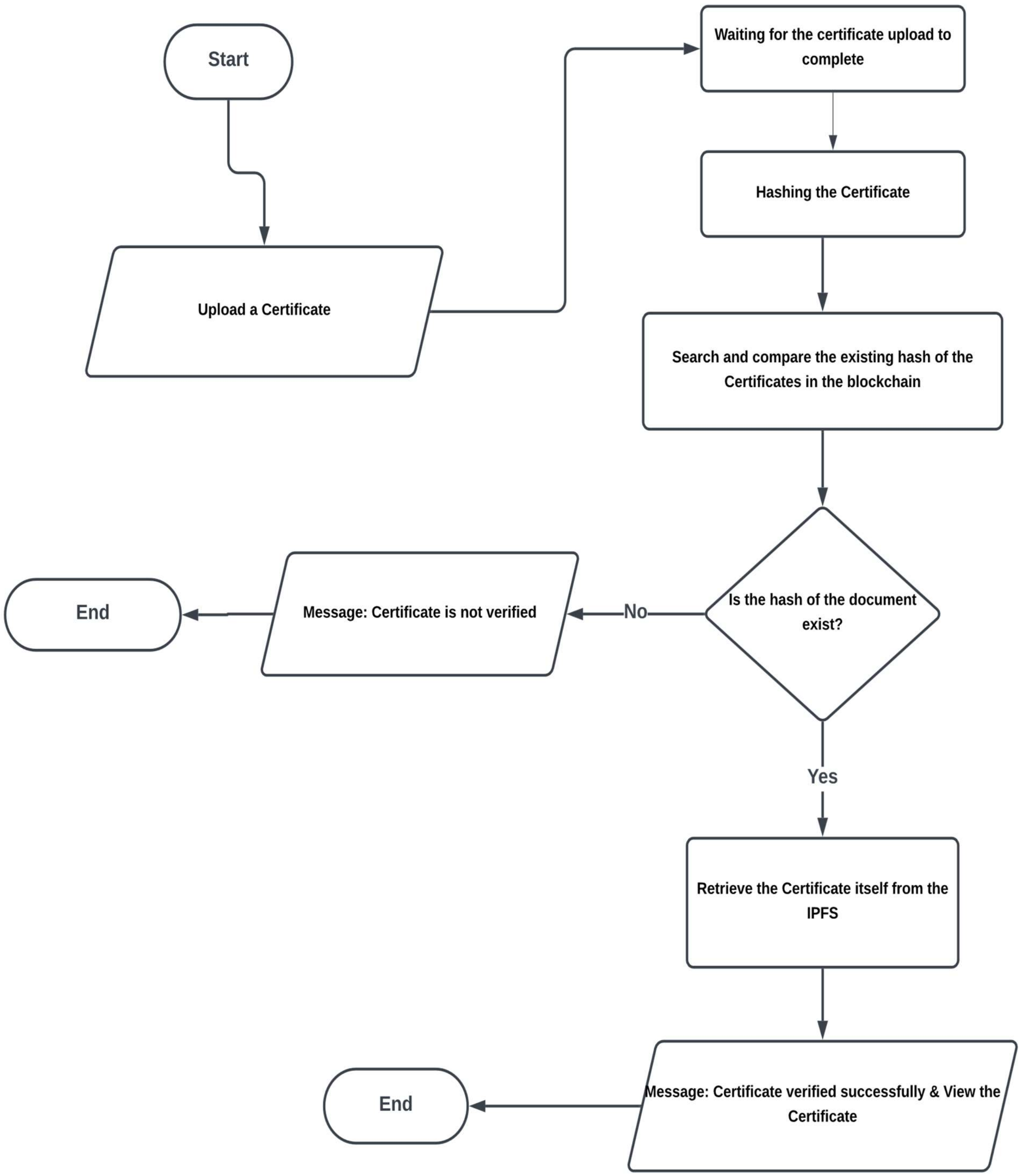

Figure 3.0.4 Flowchart of the process of verifying a Certificate

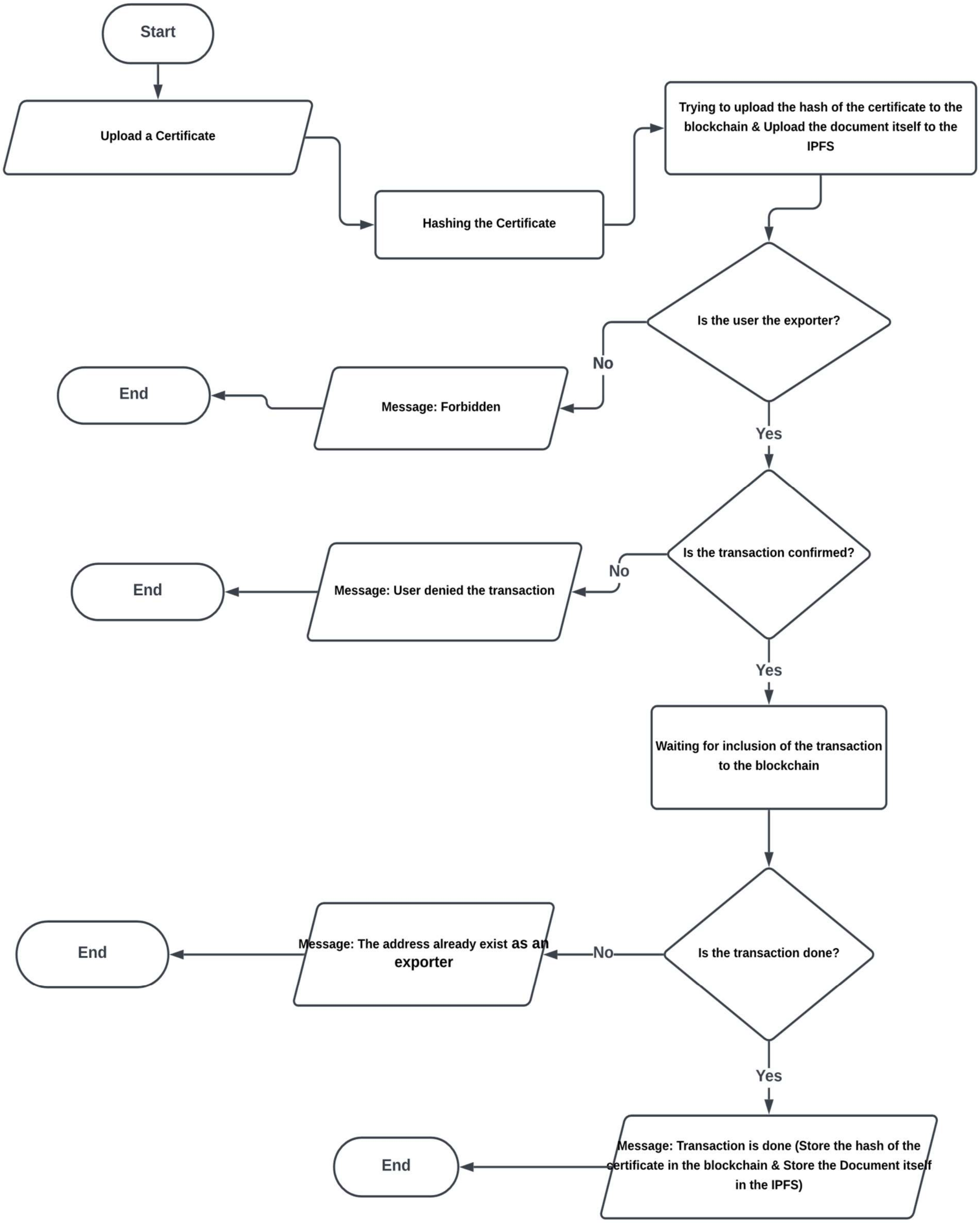

Figure 3.0.5 Flowchart of the process of Issuing Certificate

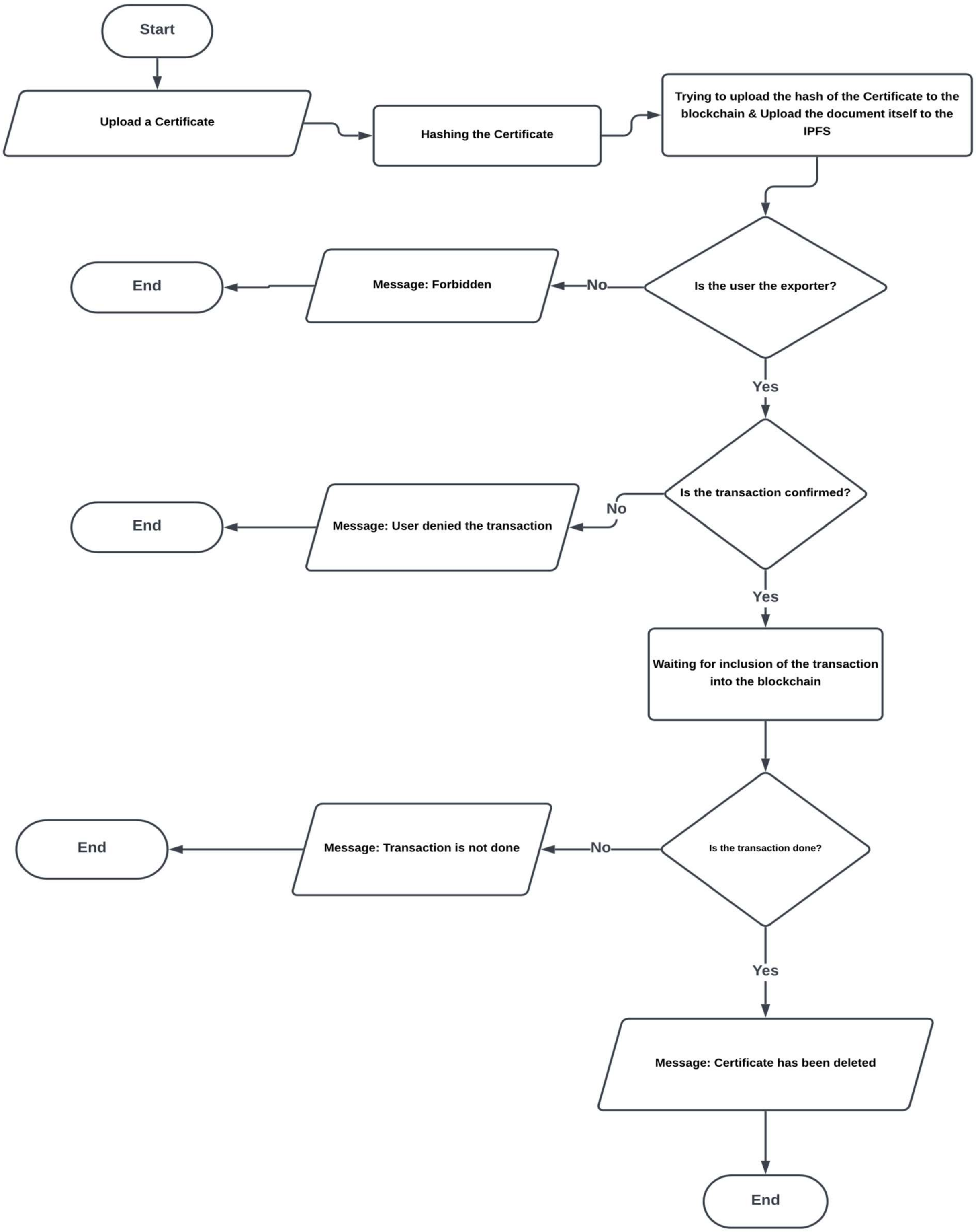

Figure 3.0.6 Flowchart of the Process of Revoking a Certificate

3.5 DEVELOPMENT METHODOLOGY

Because the Agile methodology was fast, iterative, and flexible, it could immediately respond to changes and improvements throughout the development process, making it a perfect fit for this project. Creating a blockchain-based certificate issuance and verification system for Nigerian universities was a meticulous process requiring components like integrating blockchain, developing smart contracts, and securing storage on IPFS. The tasks were managed and coordinated using Agile methodology.

I was able to implement minor iterations and test small units of the system at a time. This incremental approach ensured everything would eventually work out to be on track and on schedule in meeting milestones and deadlines. Each iteration was followed by extensive testing to uncover and fix defects as soon as they appeared during the software development lifecycle.

Agile methodology facilitated adaptability in the event of changing requirements or risks. During integration, this also meant I could easily move back and forth and adjust things without disturbing the overall project delivery. For example, creating focused sprints with clear goals and deliverables helped achieve parts of the architectural design, such as the Frontend development, Metamask wallet authentication integration, smart contract implementation, or IPFS integration for certificate storage. By developing for its users, the platform is easy to use and functional for its intended purpose.

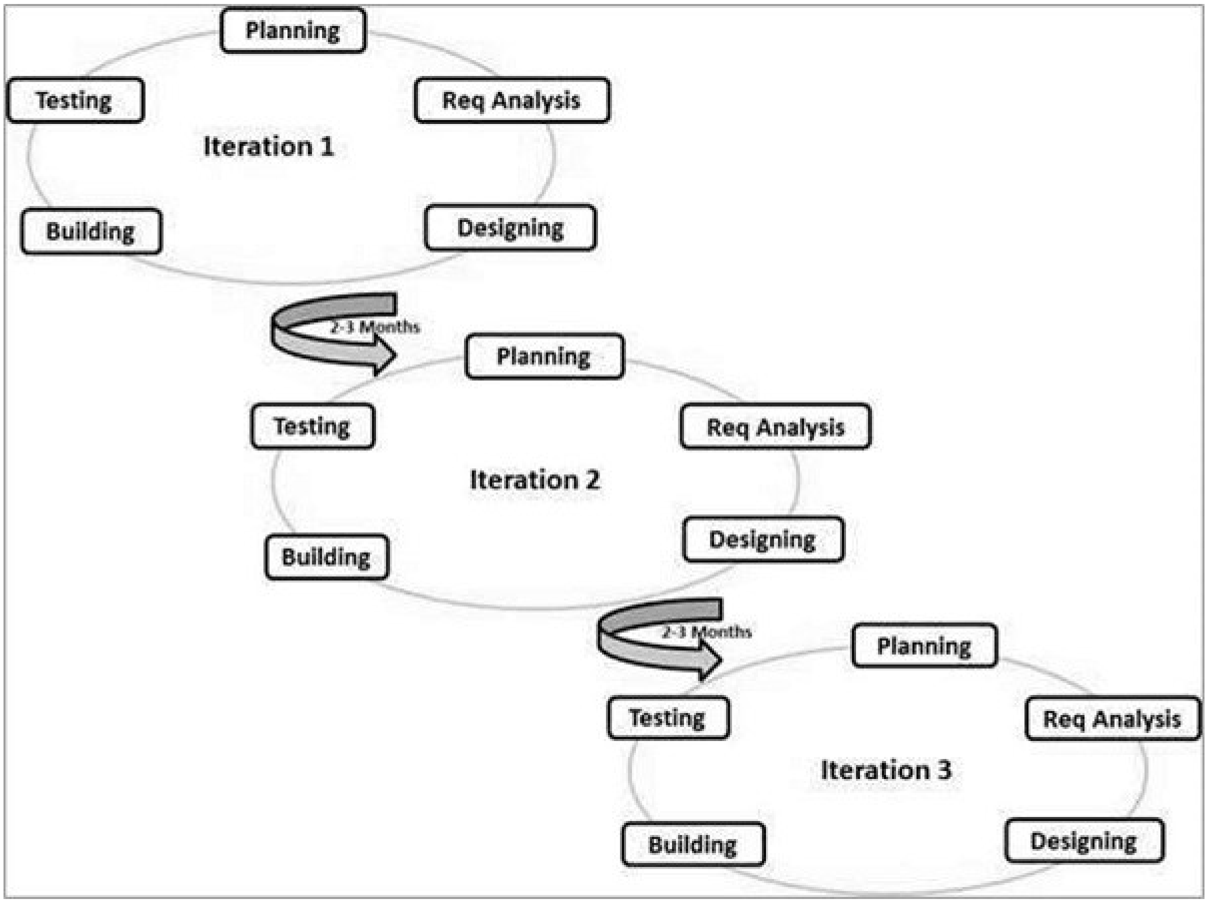

Figure 3.0.7 Agile Methodology Processes (TutorialsPoint, 2024)

3.6 SYSTEM ARCHITECTURE

Below is a picture of the system architecture and the workflow mechanism. This system involves the university that issued the document, the graduating student, and the company or institution the student aims to join for employment or postgraduate studies. Upon graduating, student certificates are stored in the NUC's blockchain-based archiving system, and graduates receive a PDF file or a QR code for validation.

When a student applies for a job or postgraduate studies, they provide the institution with their PDF or QR code file. This institution then visits the Cert Repo website, where the certificate was issued, to check the graduate student's credentials. The certificate is either verified or rejected by the issuing university.

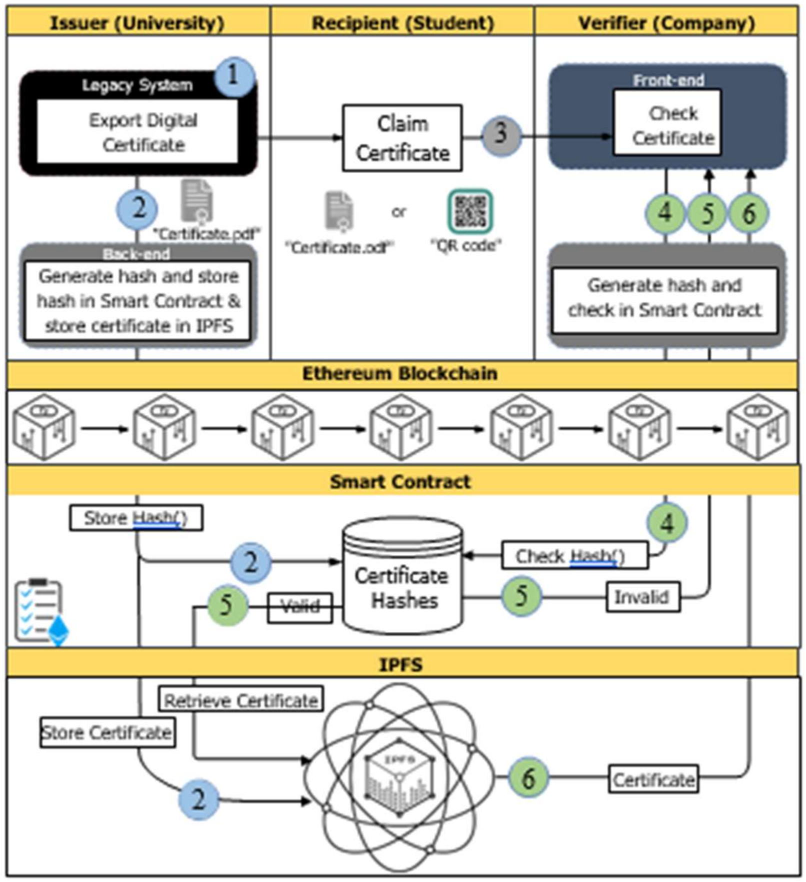

Figure 3.0.8 Architecture of Cert Repo Platform (Ali, 2022)

CHAPTER FOUR

IMPLEMENTATION, RESULT AND DISCUSSION

4.1 INTRODUCTION

This chapter describes the implementation environment of the proposed system referred to as “CertRepo” and represents the technologies and frameworks used. It describes each of the components used in the system and their functionality. A detailed description shows the interaction of the different elements and how they work together, elaborated with diagrams. In addition, we assess how the implementation conforms to and meets the project goals developed in previous chapters. This general description clearly explains how the proposed system was implemented and made functional.

4.2 COMPONENTS

The front interfaces of CertRepo are composed of six main pages. The home page has the NUC logo, a brief about the system and technology, and how the platform is used. The NUC page allows university management. The University upload page will enable universities to export their certificates to the blockchain in either PDF or image (JPEG, PNG) and generate a QR code. The Verify Cert page enables institutions to verify a student's certificate without a transaction on the Cert Repo platform. The Revoke Cert Page is allocated for the university issuers to revoke the certificate they previously issued. The Developer section, including student and supervisor, is on the team page. The figure below shows the structure of the front interfaces of the system.

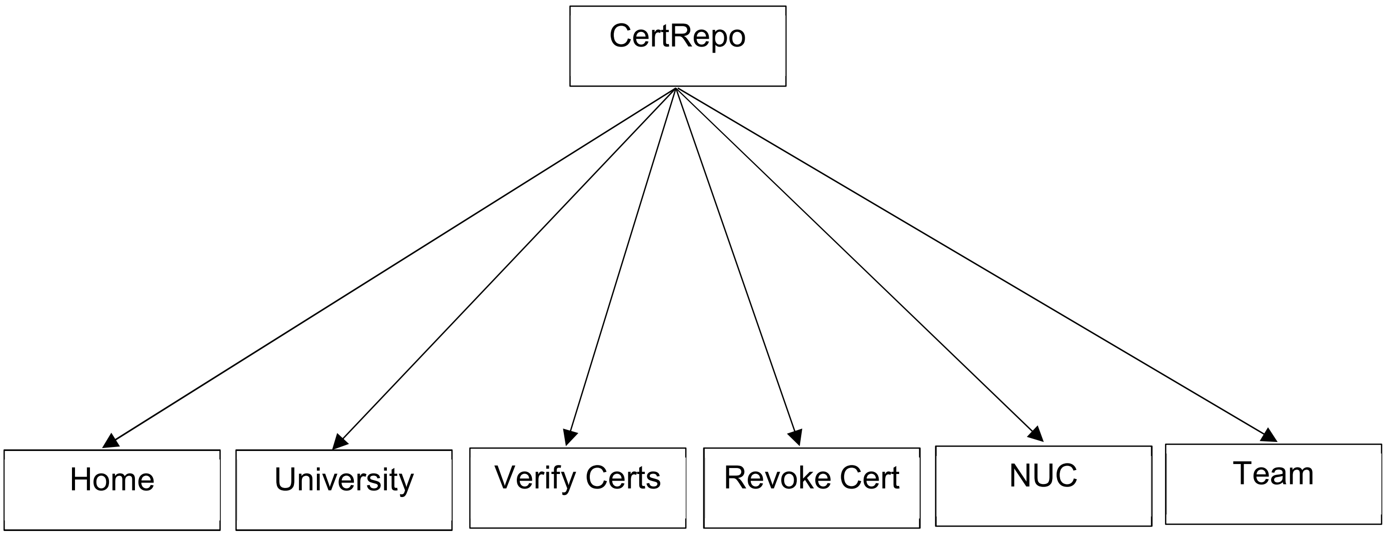

Figure 4.0.1 Components of the platform

4.3 DEVELOPMENT LANGUAGES

The front-end systems were built upon HTML, CSS, JavaScript, and BootStrap. HTML is a standard Markup language being followed in designing structures for web content, further being made more manageable by the technologies of CSS and scripting languages like JavaScript, separately other than that of Bootstrap. CSS is one of the straightforward mechanisms used for web styling to permit the addition of styles such as fonts, colour, and spacing to web documents to improve its screen impact. JavaScript is a versatile and lightweight programming language that is commonly interpreted and used for a wide range of network-centric applications. It is open and cross-platform, facilitating interactivity and engagement with the user on the front end. Bootstrap is such a robust front-end framework applied in creating and designing modern sites and web apps, thus ensuring that the website created is responsive and equally accessible from any smart device, table top, or portable. Smart contracts implemented on the blockchain were written using Solidity, which is a contract-oriented programming language for implementing smart contracts on blockchain platforms. Smart contracts allow for secure and transparent execution of contracts, which is important for any functionality of and integrity in a blockchain-based system. The system was also integrated with IPFS for decentralized file storage and sharing. IPFS is a peer-to-peer protocol and network for a distributed file system that allows efficient data storage and sharing. It efficiently and securely stores data, ensuring it remains immutable and accessible across a decentralized network.

4.4 RESULT

This section is dedicated to presenting implementation results within the project, specifically to the developed features and functionalities for the blockchain-based certificate issuance and verification system: "Cert Repo." Outlined below are the functionalities and properties of Cert Repo, which were developed according to the requirements set in the objectives of this project. The results show that the system can issue, store, and securely verify certificates while at the same time proving their potential for authenticity and integrity in academic credentials.

I. Home Page

The home page provides an overview of the system, including its features and the services it offers, as depicted in the image below:

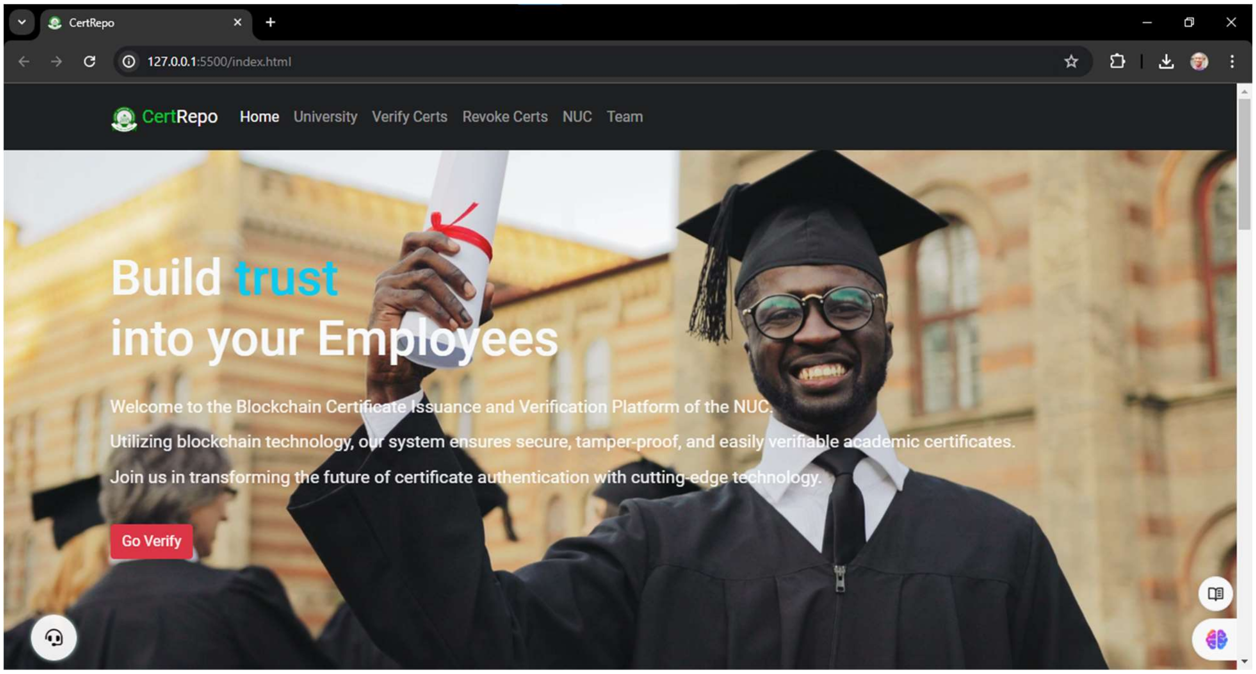

Figure 4.0.2 Home page of Cert Repo Platform

II. Team Page

The Team page contains the team responsible for the execution of this project, which includes the student and his supervisor. All team contacts are available to reach out to them, as shown in the image below:

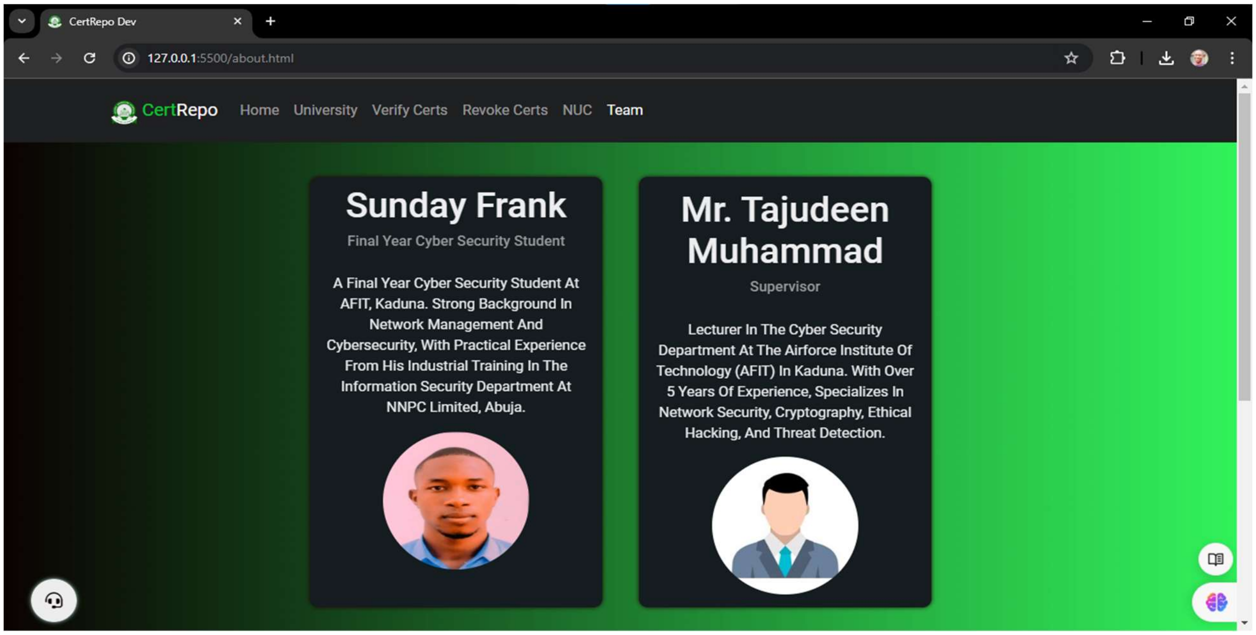

Figure 4.0.3 Team Page of Cert Repo Platform

III. NUC Page

Through the NUC page, the body who is the owner of the smart contract can add and edit a university or delete a university from the exporting certificate to the blockchain, as shown in the image below:

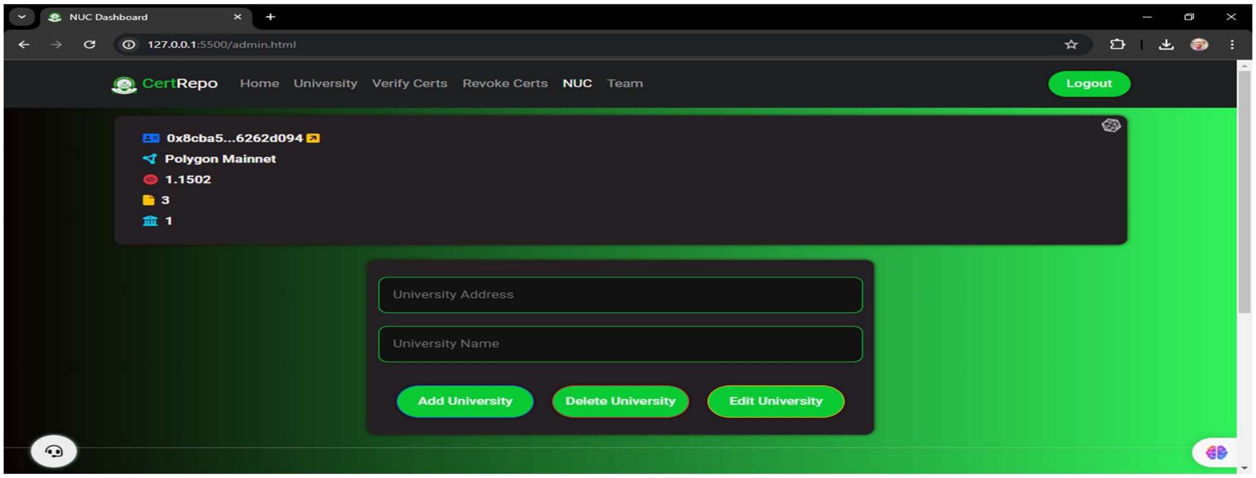

Figure 4.0.4 NUC Page of Cert Repo Platform

In the top left, various information is displayed, such as the address of the deployer/owner of the smart contract (NUC), the network name where the smart contract is deployed, the wallet balance in this account, the count of student certificates issued to the blockchain by universities, and the number of certificate exporters (Universities), as depicted in the image below:

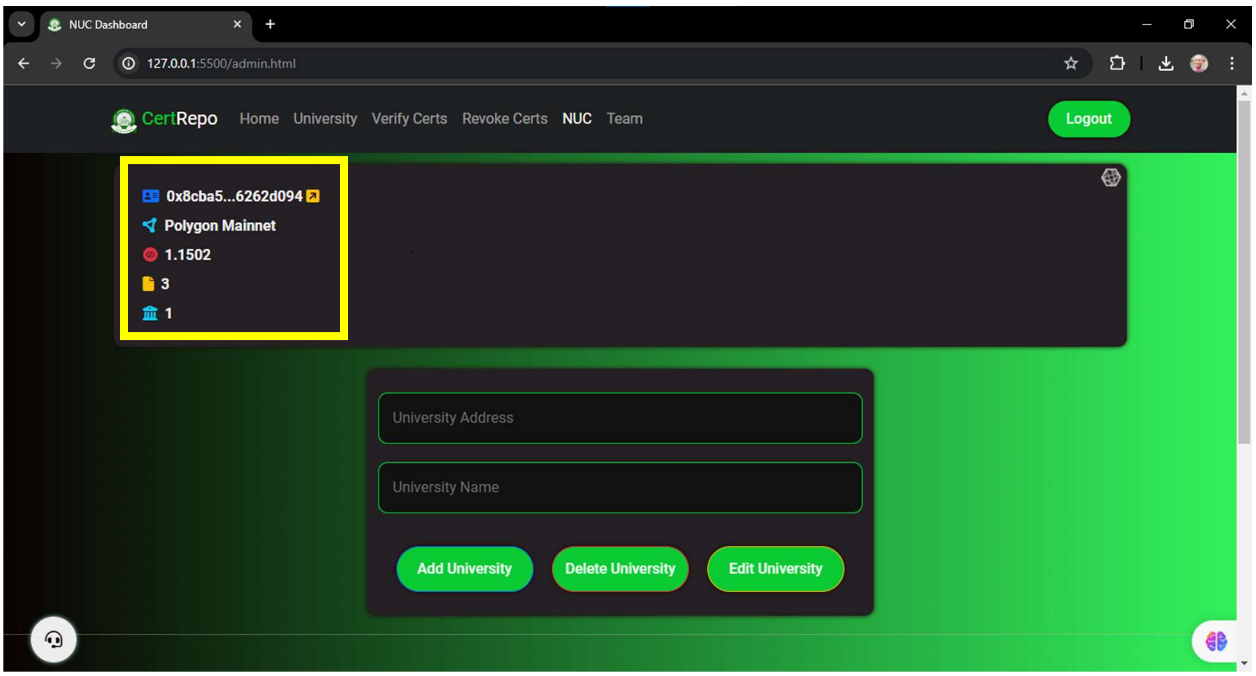

Figure 4.0.5 Properties of the NUC Page

The NUC adds a new university by including the university's address and the name of the issuing university, as depicted in the image below:

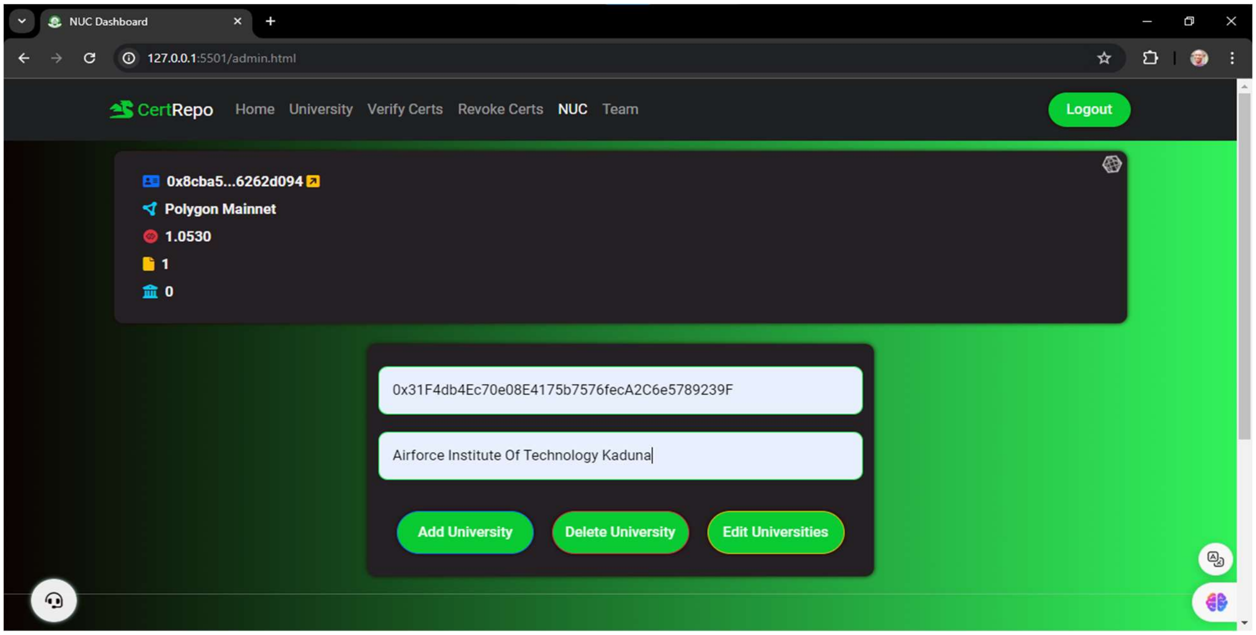

Figure 4.0.6 NUC's Page "Add University" Operation

After clicking the "Add University" button, it will wait for your confirmation or rejection of the transaction, as depicted in the image below:

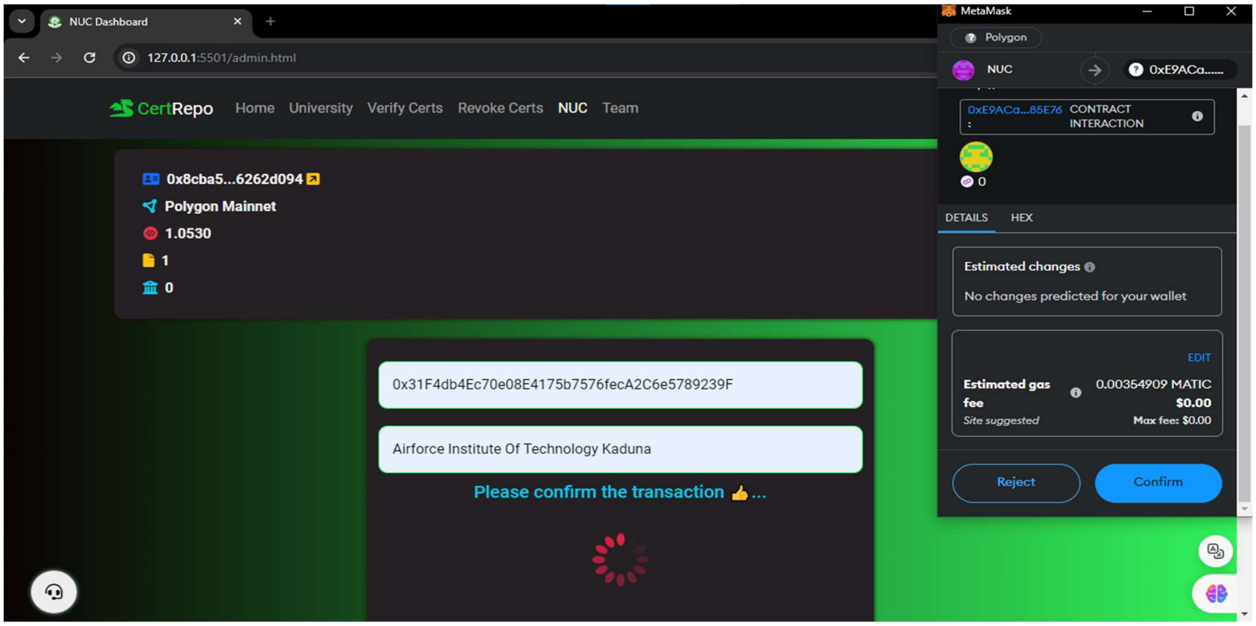

Figure 4.0.7 Confirmation of "Add University" operation

After confirming the transaction by pressing the "Confirm" button in Metamask, the process of adding a certificate issuer will be successfully completed, as depicted in the image below:

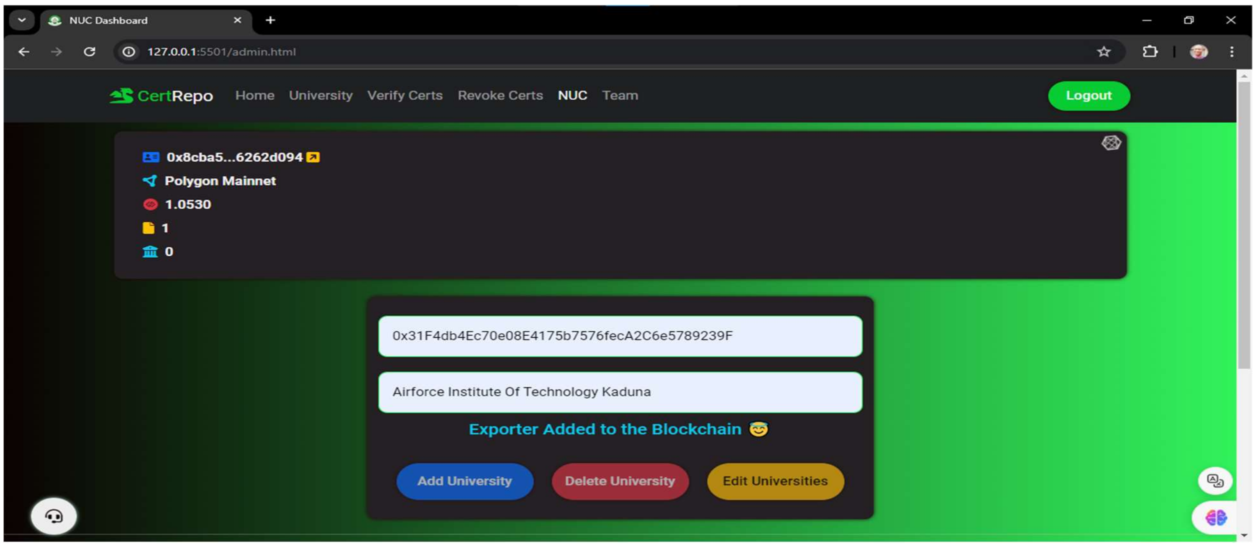

Figure 4.0.8 Successful Deletion of a Certificate Exporter

The process of editing or removing a certificate exporter is carried out by following the same steps as adding a university exporter above.

IV. University Page

The university page focuses on exporting certificates from universities. Each certificate exporter uploads graduating students' certificates from that university, as seen in the image below:

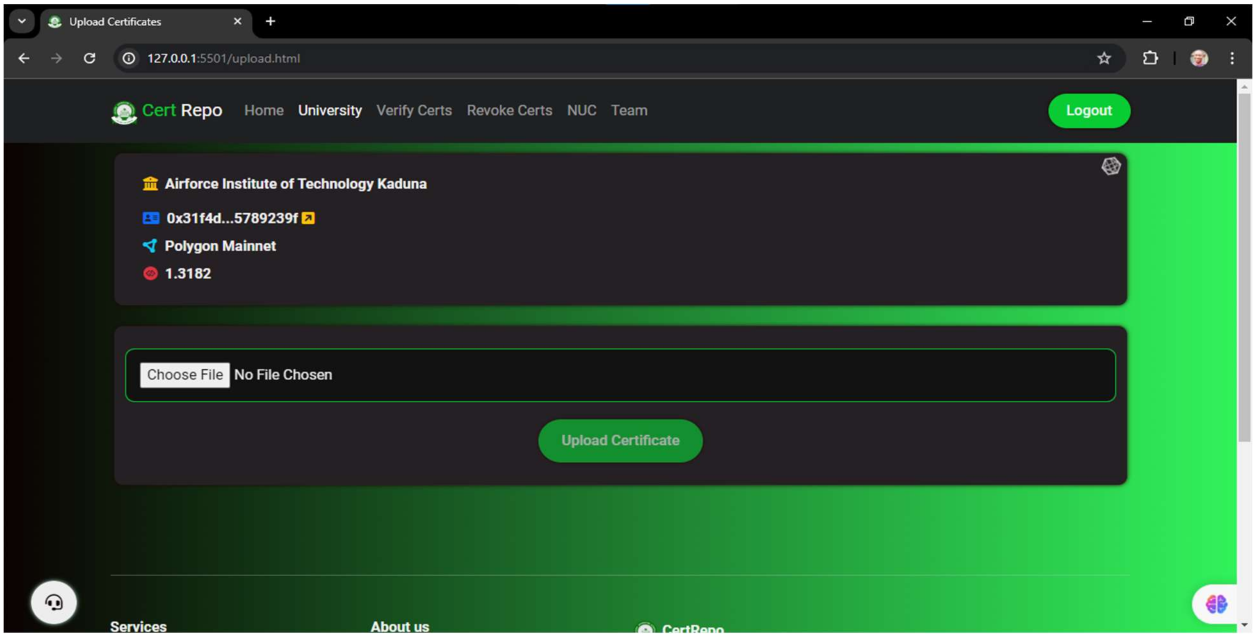

Figure 4.0.9 Interface of the University page

In the top left corner, various details are displayed: the university's name issuing the certificate, the exporter's address, the network name where the smart contract is published, and the wallet balance in this account, all visible in the image below:

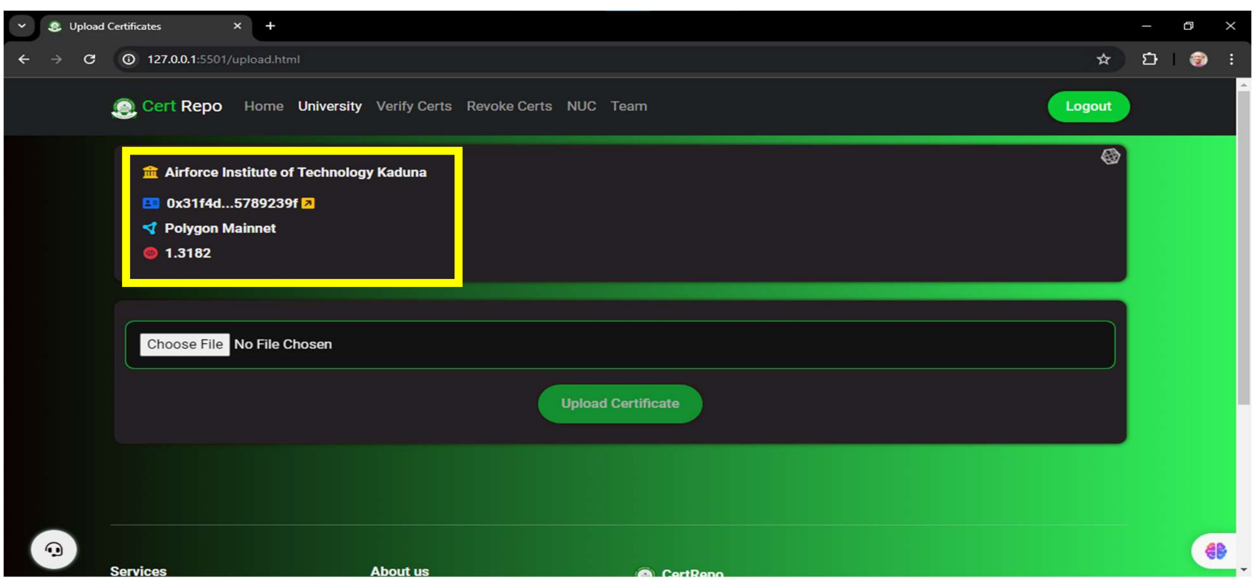

Figure 4.0.10 University Details on the University Page

The certificate uploading process involves selecting the certificate file using the "Choose File" option. Subsequently, the certificate undergoes direct hashing, as depicted in the image below:

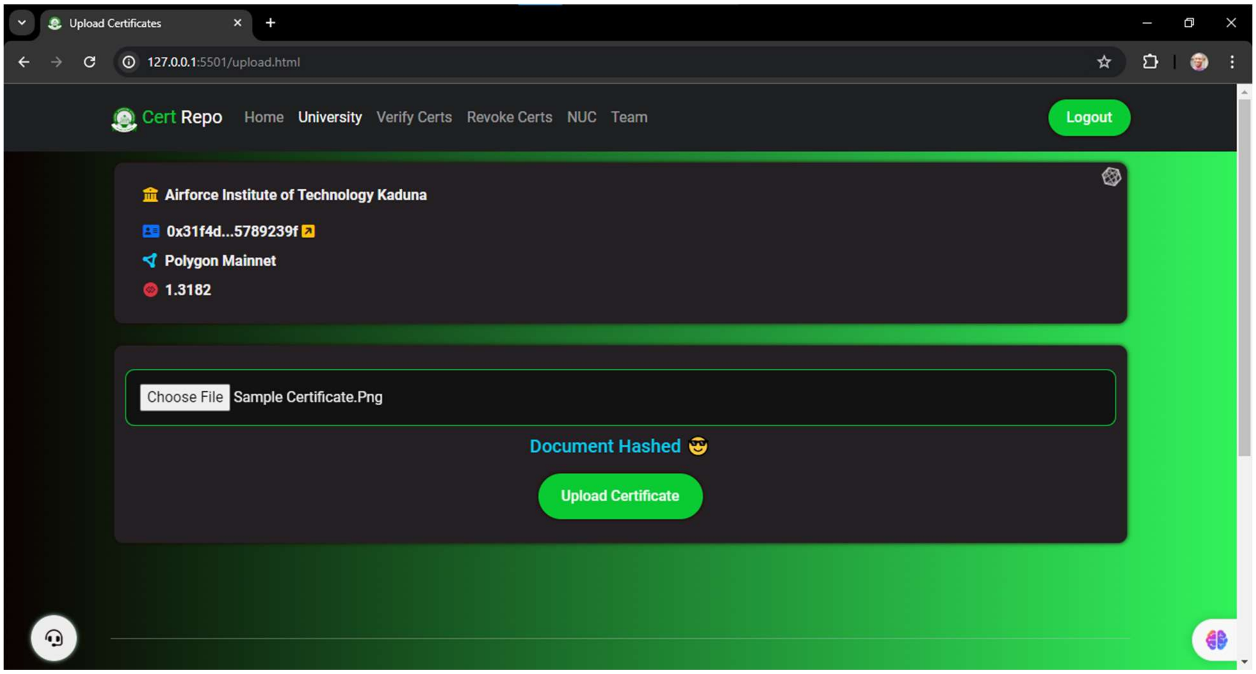

Figure 4.0.11 Showing a hashed sample certificate

After clicking the "Upload Certificate" button, it will wait for transaction confirmation or rejection, as depicted in the image below:

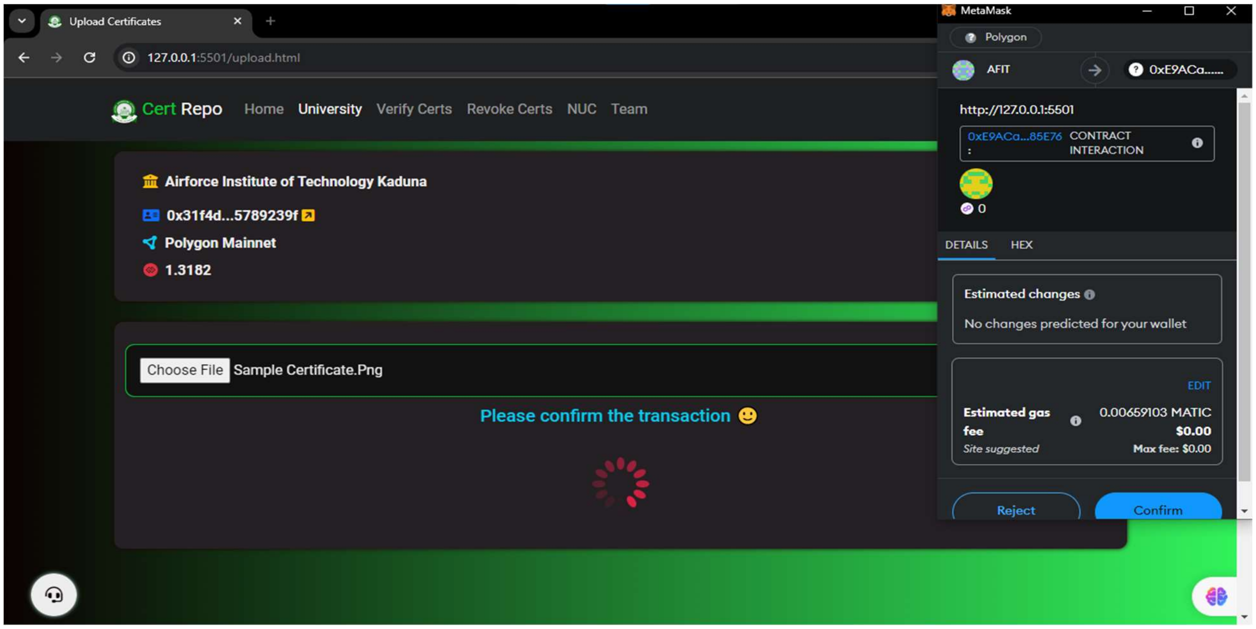

Figure 4.0.12 Showing Confirmation of Certificate Issuing Transaction

After pressing "Confirm," the student's certificate will be uploaded to IPFS successfully, as shown in the images below.

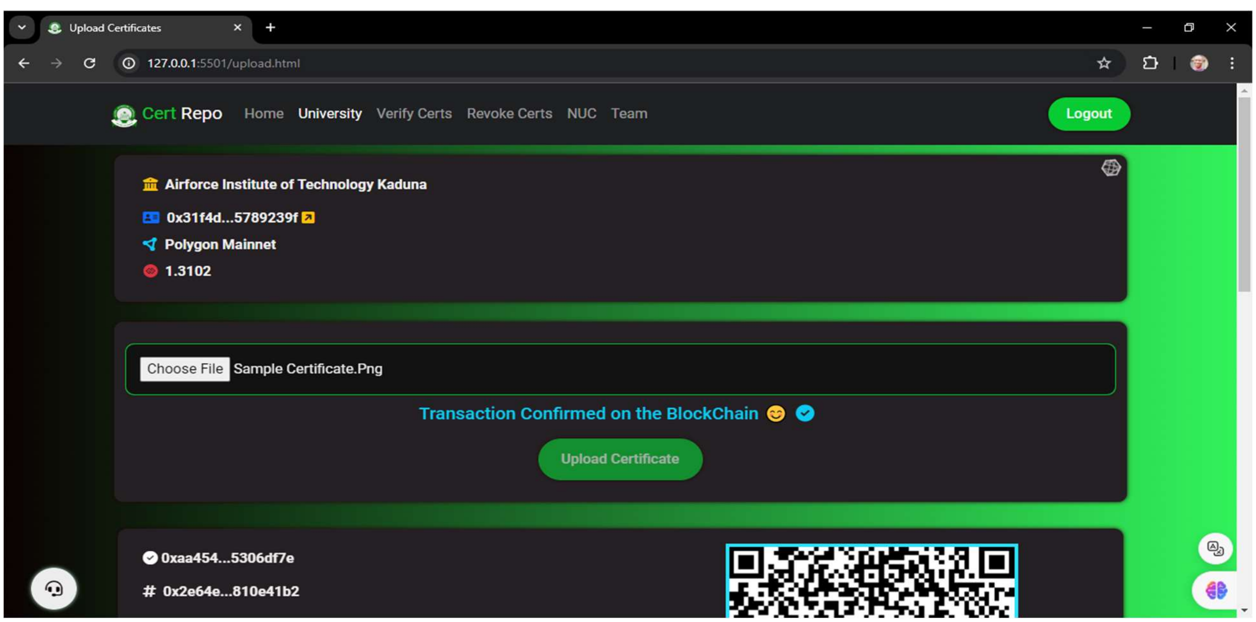

Figure 4.0.13 Showing a Successful Certificate Upload to IPFS

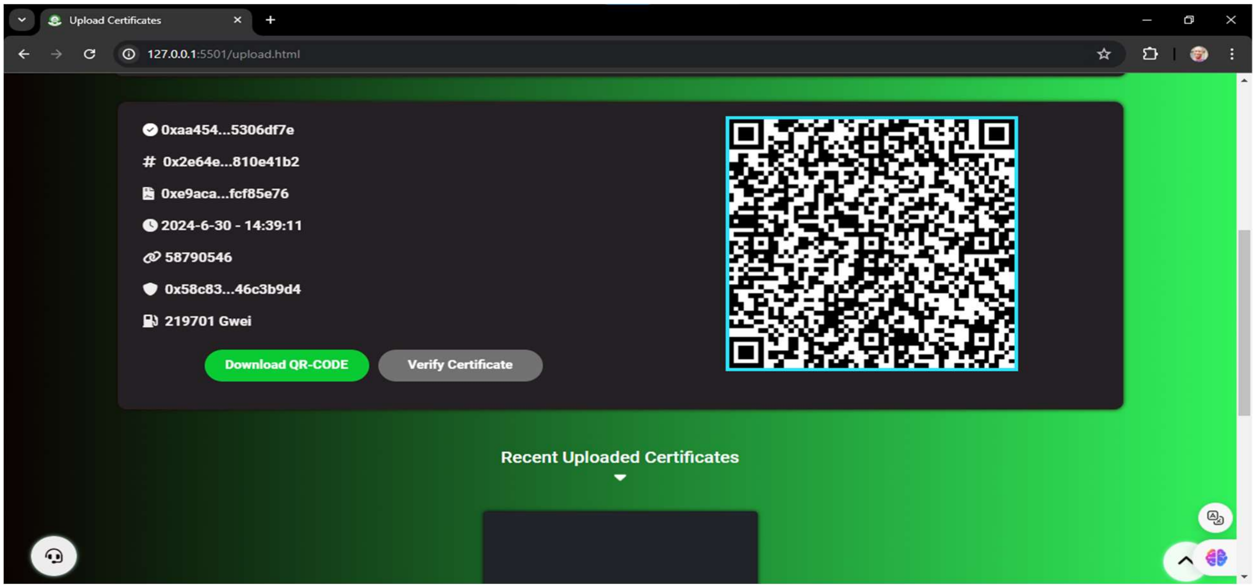

Figure 4.0.14 Showing Recent Uploaded Certificate by the University

After the upload is finished, a series of details emerge, such as the issuing university's address, the completion time, and the blockchain's block number where the transaction was added. Additionally, a QR code is displayed for student verification, as depicted in the image below:

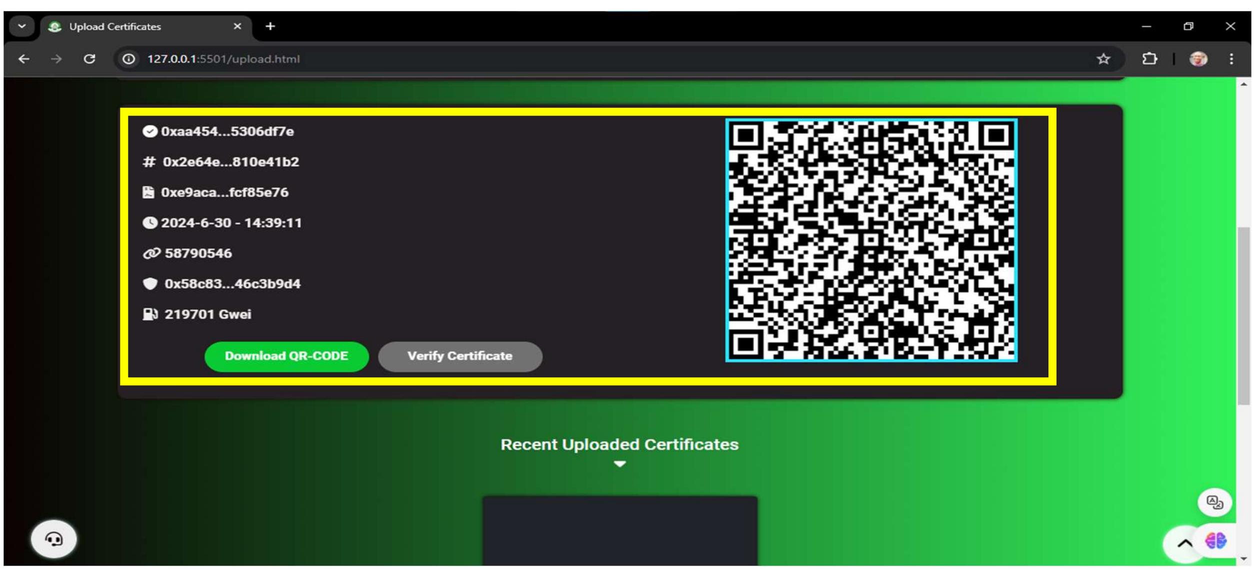

Figure 4.0.15 Showing Details of an Issued Certificate

Most recently issued student certificate are shown for each university as shown in the image above.

After executing an upload process and incorporating a certificate into the blockchain network, we observe a little reduction in the wallet's balance, highlighting the application of transaction fees, as depicted in the image below:

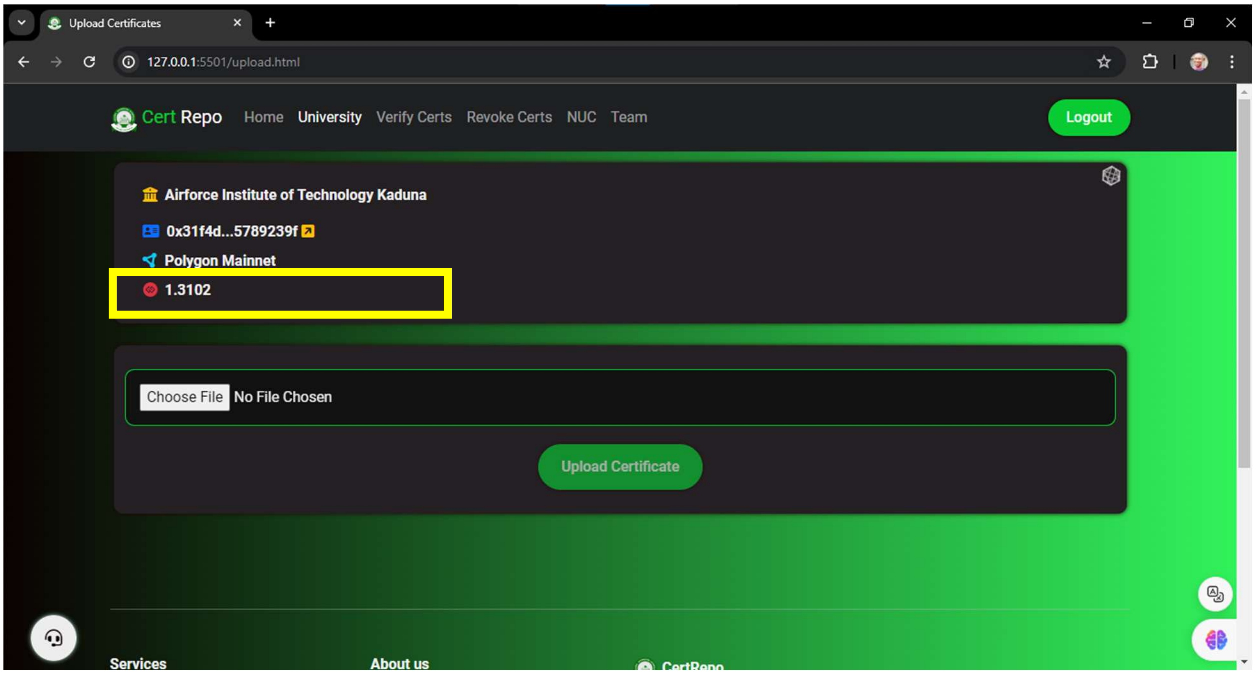

Figure 4.0.16 Showing Wallet Reduction After an Upload Operation

V. **Verify Certs Page**

After completing their studies, the student aims to secure employment or pursue postgraduate education. To do so, the student provides his assigned QR code and academic credentials to the relevant institution. It is essential for organizations to authenticate the applicant's credentials. This verification process involves accessing the NUC's Cert Repo platform on the official website. The verification page can be found on the Cert Repo platform, depicted in the image below:

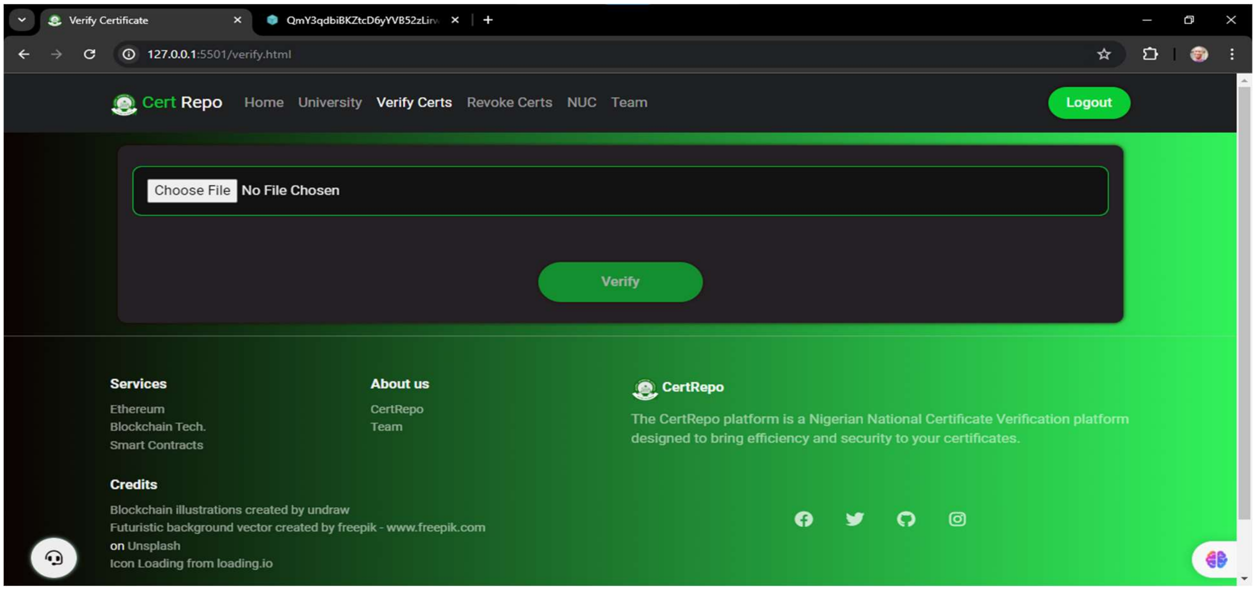

Figure 4.0.17 Showing the Verify Certs Page

After the certificate file is uploaded (or via a QR code), the certificate will be hashed, as depicted in the image below:

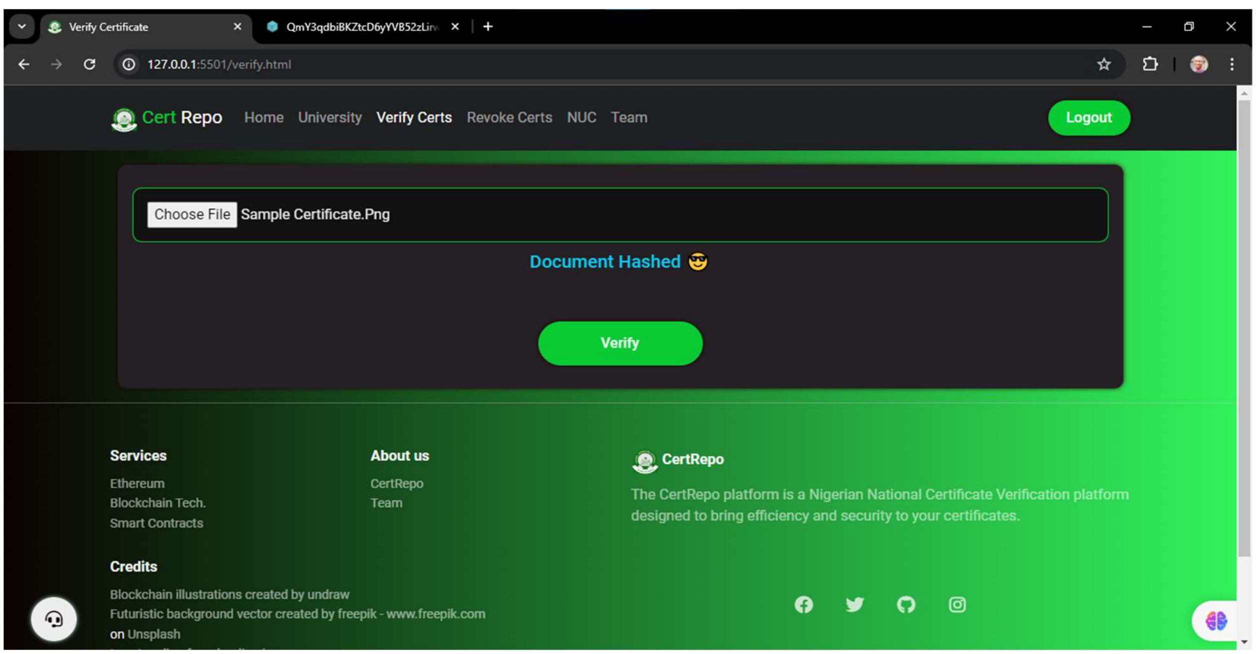

Figure 4.0.18 Showing a Hashed Certificate File for Verification

Upon pressing the "Verify" button, the system will compare the file's hash with those previously stored in the blockchain. If a match is found, the system will access IPFS to retrieve the certificate. Consequently, the verification request will be confirmed, and the certificate will be presented, as illustrated in the image below:

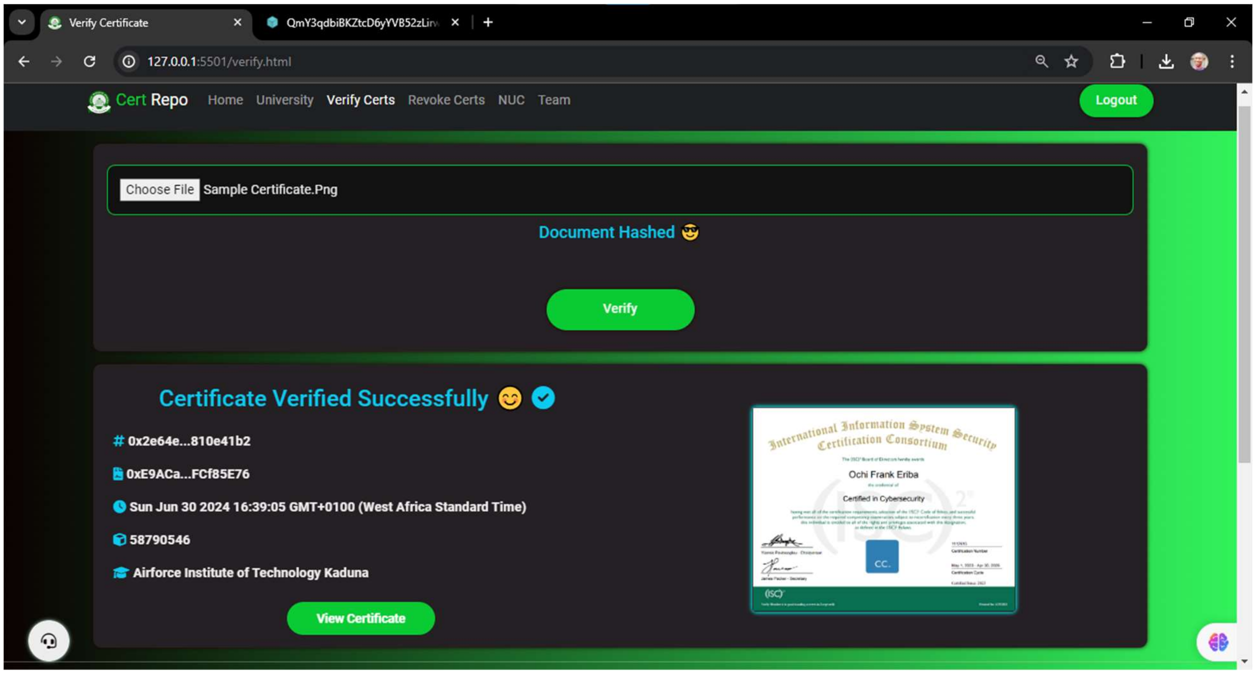

Figure 4.0.19 Showing a Verified Certificate

The simplified verification workflow is shown in the illustration below:

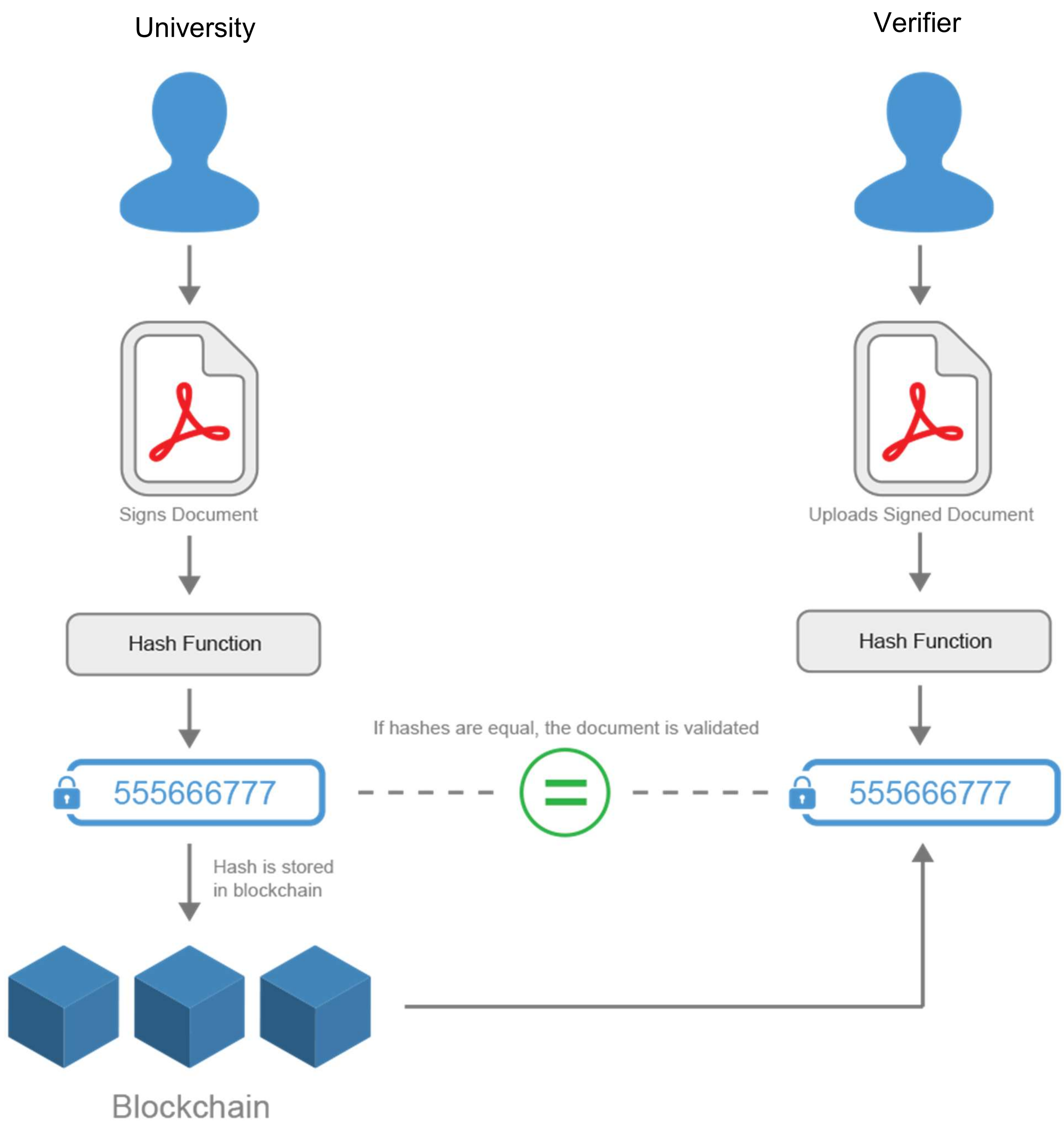

Figure 4.0.20 Showing Certificate Verification Workflow (Arlanto, 2024)

If the file hash does not correspond to any of the hashed files stored in the blockchain, the verification request will be denied, as depicted in the image below:

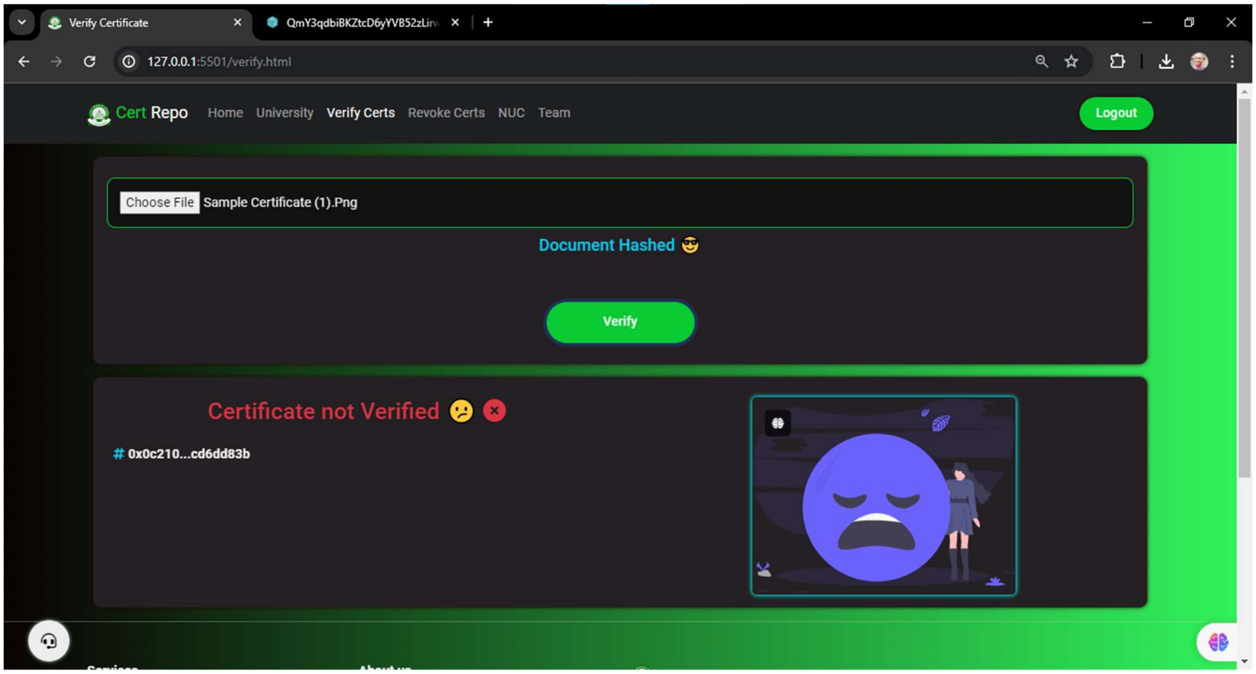

Figure 4.0.21 Showing a Fake Certificate Identification

The verification function is not executed as a transaction through the MetaMask wallet; instead, it is performed via a dedicated node.

VI. Revoke Certs Page

In Nigerian universities, a certificate can be revoked if it is found to have been wrongly awarded to a student or if an error requiring correction is discovered. Consequently, university authorities have the power to annul a previously issued certificate. Access to this page is limited to universities only. The certificate revocation page is illustrated in the image below:

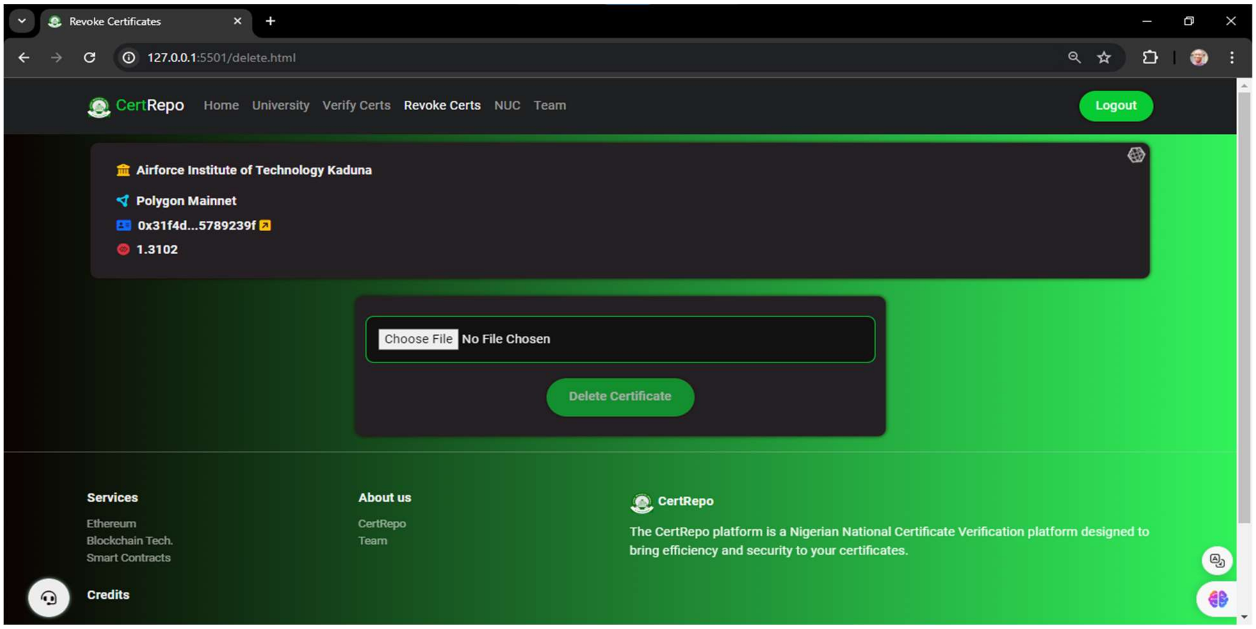

Figure 4.0.22 Showing Revoke Certificates Page

After uploading the certificate file, it will be hashed as displayed in the image below:

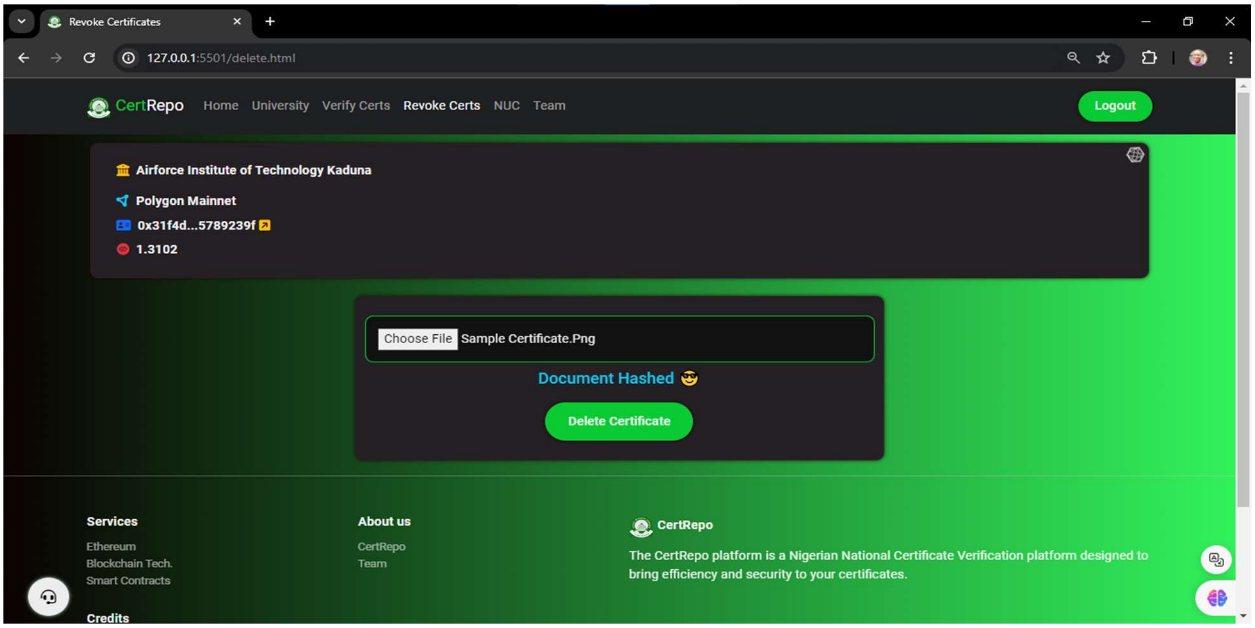

Figure 4.0.23 Showing a Selected Certificate File for Revocation

Upon selecting the 'Delete Certificate' button, a prompt will appear to either validate or decline the action, as illustrated in the image shown:

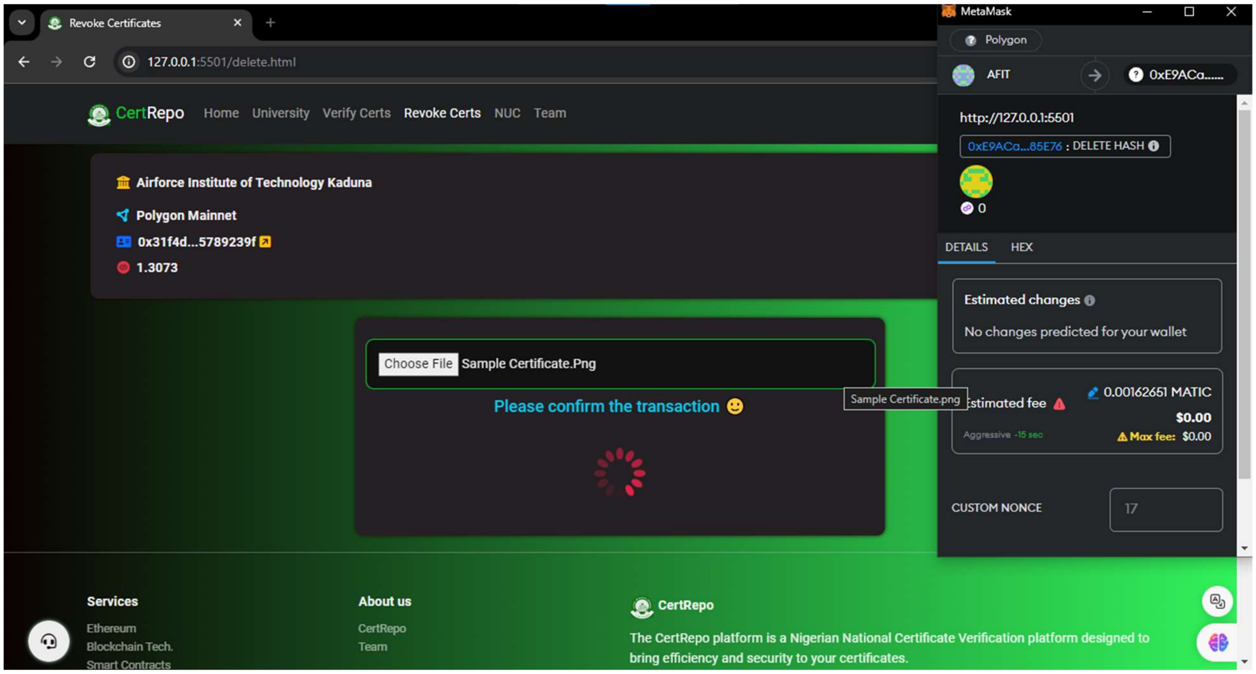

Figure 4.0.24 Showing Certificate Revocation Transaction Confirmation

If the transaction is confirmed by clicking the "Confirm" button on Metamask, the process of revoking a student's certificate will be successfully completed, as depicted in the image below:

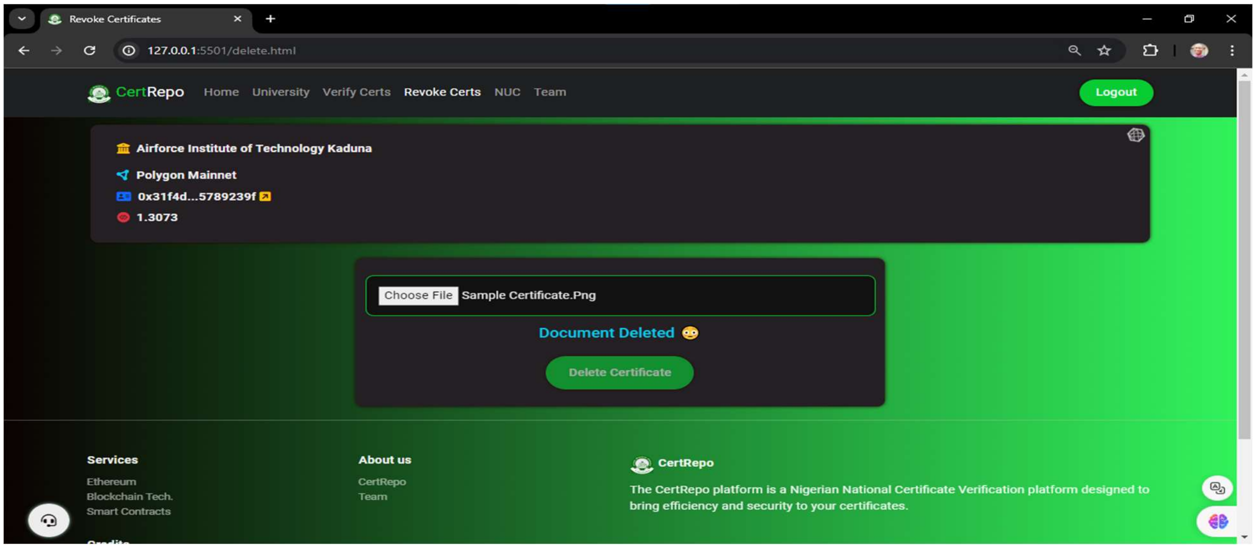

Figure 4.0.25 Showing a Successful Revoked Certificate

We can now observe that the wallet's balance has dropped a little following the removal of a certificate from the blockchain network. This is illustrated in the image below, showing the application of fees for every transaction:

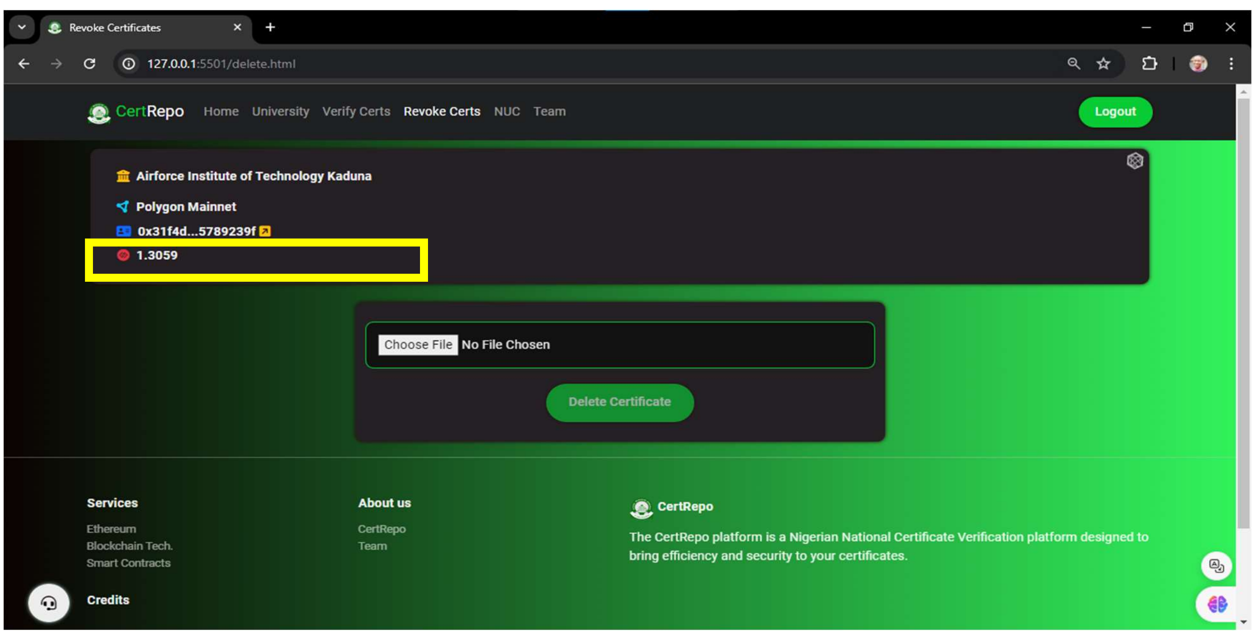

Figure 4.0.26 Showing a Reduced Wallet Balance After a Revocation Operation

4.5 EVALUATION OF CERTREPO IN COMPARISON TO THE BASE SYSTEM

The evaluation of CertRepo, deployed on Polygon, against the base system on Binance Smart Chain (BSC) reveals significant enhancements across critical performance metrics. These improvements highlight the distinct advantages CertRepo offers, positioning it as a more efficient and scalable solution for academic certificate management.

I. **Transaction Costs:** On Binance Smart Chain, the cost to issue or revoke a certificate averages $0.1 (₦165.54), especially during periods of low network activity (Hassan & Salim, 2022). In contrast, CertRepo on Polygon reduces this significantly, with transactions costing on average just 0.00354909 MATIC, approximately $0.003 (₦4.97). This reduction is critical for large-scale operations, as CertRepo is designed to handle thousands or millions of certificates. Polygon's Layer 2 scaling solution enables CertRepo to process high transaction volumes while maintaining cost-efficiency, making it an ideal choice for institutions dealing with large certificate management demands (Nailwal & Kanani, 2021).

II. **Transaction Speed**: The base system on BSC processes about 100 to 300 transactions per second (TPS), with confirmation times averaging 3 seconds (Effiong, 2020). CertRepo, however, capitalizes on Polygon's capability to handle up to 65,000 TPS with confirmation times ranging from 2 to 4 seconds (Benet, 2019). This improved speed is crucial for CertRepo, where real-time issuance and verification of certificates are essential to maintaining seamless operations across multiple universities. The adoption of Polygon ensures a faster, more reliable experience for both issuers and verifiers.

III. **Scalability:** While Binance Smart Chain scales vertically, which can limit flexibility, CertRepo uses Polygon's sidechain architecture for horizontal scalability. This allows the platform to expand easily by adding additional sidechains as necessary, accommodating increasing loads as more universities join the system (Benet, 2019). This design makes CertRepo highly scalable, enabling it to handle millions of users and certificates without degrading performance.

IV. **Security:** Given CertRepo's handling of sensitive academic data, security is paramount. Binance Smart Chain's Proof-of-Stake Authority (PoSA) mechanism, with fewer validators, is more centralized, raising potential security concerns (Hassan & Salim, 2022). In contrast, CertRepo leverages Polygon's robust security model, which benefits from Ethereum's decentralized Proof-of-Stake framework (Buterin, 2020). This enhanced security ensures higher data integrity and trustworthiness, which is critical for managing academic certifications.

V. **Support for Multiple Universities:** Perhaps one of the most significant advantages of CertRepo is its ability to serve all Nigerian universities. While the traditional paper-based system is often restricted to individual institutions and their departments (Omijeh & Okeke, 2023), CertRepo supports nationwide deployment. This system allows multiple institutions to issue and verify certificates through a unified, decentralized platform. This scalability simplifies the verification process for employers and educational institutions, making CertRepo a far-reaching, efficient solution for Nigeria's academic landscape.

CertRepo significantly outperforms the base system on Binance Smart Chain in all aspects, including transaction costs, speed, scalability, ecosystem integration, security, and broad support for Nigerian universities. These advantages position CertRepo as the ideal platform for issuing and verifying academic certificates, offering a secure, scalable, and cost-effective solution for Nigeria's educational institutions.

CHAPTER FIVE

CONCLUSIONS AND RECOMMENDATIONS

This chapter contains a detailed overview of the research project which gives an unbiased account of some main results offering conclusions, managerial impact or recommendations and tying up loose ends like reflections, etc. This systematic method makes sure that the groundwork of how big this study can affect and improve.

5.1 SUMMARY OF KEY FINDINGS

Their project revolved around the development and implementation of a blockchain certificate issuing/verification system dubbed "Cert Repo" to be used by Nigerian Universities. To improve the security, efficiency, and reliability of academic credentials was one of its principal objectives. Key takeaways are:

I. **System Development Success:** The smart contracts were finalized and developed in Solidity, the decentralized storage was implemented on IPFS, with backend code utilizing a variety of technologies including those required for web development. Therefore, it has minted the academic credentials which in return issued to be saved and verified with more than enough cost-effective.

II. **Security:** By its very nature, blockchain ensures that the information stored is safely retained and hard to replicate which reduces the risk of certificate forgery. No certificate could be changed or copied without recognition.

III. **Encouraging User Feedback:** From the trial rollout, feedback has been positive with high user satisfaction in both usability and reliability of the system. University administrators experienced the certificate issuance and verification as refined and secure.

IV. **Cost:** The cost to set up was high, but the improvements in the long term resulted in less administrative work and hence more secure.

5.2 CONCLUSIONS

The Cert Repo blockchain technology solution is designed to be used for the issuance and verification of certificates in Nigerian universities. It increases the reliability and scalability of a national-level rollout by securing the academic credential process. The research presented ultimately suggests that blockchain technology is a theoretically legitimate and practically adequate solution for the reliable protection of academic credentials, which simultaneously lowers significant administrative costs.

5.3 PRACTICAL IMPLICATIONS

Educational sector practical implications of the study:

I. **Security:** by Leveraging the Cert Repo system, Cert Repo provides a security guarantee for academic certificates to prevent fraud and maintain the identity of educational institutions.

II. **Operational Efficiency:** Automating the process of issuance and verification not only reduces workload but also increases efficiency - for institutions as well as students.

III. **Improved Trust:** With the openness of blockchain technology, trust will be flowering between employers and students as well as educational institutions.

5.4 RECOMMENDATION FOR FUTURE WORKS

Implications for Future Research — Ideas about What Researchers Should Study Next

I. **Scalability Considerations:** Research how to make the Cert Repo system more scalable so that it can serve not only additional institutions and higher numbers of certificates.

II. **Minimize Cost:** Use another blockchain if possible to lower transaction costs during setup or explore ways of optimizing the logic and requirements as described in Step 2.

III. **Expand Use Cases:** The blockchain may be extended to other categories of academic information and further links with international verification systems are on the cards.

5.5 REFLECTIONS

Looking back at the project, developing and implementing the Cert Repo system showed how blockchain technology can bring positive changes to education. This lesson drove home the necessity of reconciling technical and operational difficulties as organizations attempt to leverage new forms of technological innovation.

5.6 PRACTICE RECOMMENDATIONS

Practical Suggestions for Success Ensuring Cert Repo has an impact can be achieved by emulating the following approach set:

I. **Standardized Training:** standardize the introduction of this new system through legitimate and effective use in university administrators and staff.

II. **Policy Development:** Formulate institutional policies for the incorporation and use of blockchain technology in administrative related matters.

III. **Collaborative Undertakings:** Fosters an ecosystem in which both universities and technology providers work together to further develop, enhance the blockchain-based system.

IV. Create awareness campaigns to educate stakeholders on the value of Cert Repos and increase acceptance/trust in the technology.

REFERENCES

Alam, S. (2021). A blockchain-based framework for secure educational credentials. Turkish Journal of Computer and Mathematics Education (TURCOMAT).

Ali, S. I., Farouk, H., & Sharaf, H. (2022). A blockchain-based model for student information systems. Egyptian Informatics Journal.

Androulaki, E., et al. (2019). Hyperledger Fabric: A Distributed Operating System for Permissioned Blockchains. Proceedings of the Thirteenth EuroSys Conference.

Antonopoulos, A. M. (2021). Mastering Bitcoin: Unlocking Digital Cryptocurrencies. O'Reilly Media.

Bach, L. M., Mihaljevic, B., & Zagar, M. (2020). Comparative analysis of blockchain consensus algorithms.

Benet, J. (2019). IPFS Content Addressed, Versioned, P2P File System. IPFS White Paper. Retrieved from https://ipfs.io

Brown, R. G., & Carlyle, J. (2019). Corda: An Introduction. R3 Corda Technical White Paper.

Buterin, V. (2020). Ethereum: A Next-Generation Smart Contract and Decentralized Application Platform. Ethereum White Paper.

Capece, G., Levialdi Ghiron, N., & Pasquale, F. (2020). Blockchain technology: Redefining trust for digital certificates. Sustainability, 12(21), 8952.

Chan, W. K., Lee, L. H., & Fong, S. (2021). Blockchain and its applications in the media industry. Journal of Industrial Information Integration.

Chen, S., et al. (2019). Blockchain applications in healthcare and the opportunities. Journal of Medical Internet Research, 21(3), e12439.

Chivu, R. G., Popa, I. C., Orzan, M. C., Marinescu, C., Florescu, M. S., & Orzan, A. O. (2022). The role of blockchain technologies in the sustainable development of students' learning process. Sustainability.

Choi, H., Kim, Y., & Cho, H. (2019). Blockchain-based electronic voting system. The Journal of Information Technology & Systems, 38(1), 57-65.

Consensys. (2020). Remix - Ethereum IDE. Retrieved from https://consensys.net

Dannen, C. (2019). Introducing Ethereum and Solidity: Foundations of Cryptocurrency and Blockchain Programming for Beginners. Apress.

Dinh, T. T. A., et al. (2020). Untangling blockchain: A data processing view of blockchain systems. IEEE Transactions on Knowledge and Data Engineering.

Effiong, M. E. (2020). A framework for the adoption of blockchain technology in academic certificate-verification systems: A case study of Nigeria [Master's thesis, Tallinn University of Technology]. ResearchGate.

Enoch, J. D., Omijeh, B. O., & Okeke, R. O. (2023). Development and implementation of blockchain solution to enable data security at University of Port Harcourt. European Journal of Electrical Engineering and Computer Science, 7(4).

Ethdocs. (2020). Application Binary Interface (ABI) Specification. Retrieved from https://ethdocs.org

Falah Hassan, A., & Hussein Salim, A. (2022). Blockchain-Based Document Verification System. Undergraduate thesis, University of Baghdad, College of Al-

Khwarizmi Engineering, Department of Information and Communications Engineering, Baghdad, Iraq.

Fanning, K., & Centers, D. P. (2019). Blockchain and Its Coming Impact on Financial Services. Journal of Corporate Accounting & Finance.

Geetha, S. K., Sreenithi, R., & Sreenithi, R. (2020). Educational Certificate Verification Using Blockchain Based Framework. Turkish Journal of Physiotherapy and Rehabilitation.

Gräther, W., Kolvenbach, S., Ruland, R., Schütte, J., Torres, C., & Wendland, F. (2019). Blockchain for education: lifelong learning passport. ERCIM Blockchain.

Grech, A., Sood, I., & Ariño, L. (2021). Blockchain, self-sovereign identity and digital credentials: promise versus praxis in education. Frontiers in Blockchain.

Gupta, H., & Sadoghi, M. (2020). Blockchain Transaction Processing. Foundations and Trends® in Databases.

Hargude, R., Ashutosh, G., Nawale, A., & Adsure, S. (2021). Generating E-Certificate and Validation using Blockchain. International Journal of Creative Research Thoughts (IJCRT).

Hileman, G., & Rauchs, M. (2020). Global Blockchain Benchmarking Study. Cambridge Centre for Alternative Finance.

João, V. M. (2019). Document validation using blockchain, Brazil.

Kim, S., & Lee, J. (2020). A Survey of Consensus Mechanisms in Blockchain. Journal of Information Processing Systems, 16(1), 1-18.

Koens, T., & Poll, E. (2019). What blockchain alternative do you need? Data Privacy Management, Cryptocurrencies and Blockchain Technology.

Mangaiarkarasi, T., Prem, M., Madhavan, M., Vijaya Kumar, R., & Jaisurya, J. (2020). Blockchain-based certificate validation. International Journal of Applied Engineering and Technology.

Mendling, J., Weber, I., & Van der Aalst, W. M. (2021). Blockchains for Business Process Management: Challenges and Opportunities. Springer.

Nailwal, S., & Kanani, J. (2021). Scaling Ethereum and Accelerating Web3 Development: The Story of Polygon. Polygon Network White Paper

Nakamoto, S. (2019). Bitcoin: A Peer-to-Peer Electronic Cash System. Bitcoin.org White Paper.

Omar, S. S., Ghazali, O., & Rana, M. E. (2020). Blockchain based Framework for Educational Certificates Verification. Journal of Critical Reviews (JCR).

Polygon. (2021). Polygon (Matic) Network. Retrieved from https://polygon.technology

Prata, D. N., de Araújo, H. X., & Santos, C. (2019). Blockchain Technology applied to Education. International Journal of Information Technology.

Pustejovsky, J. (2020). Monitoring Blockchain Network Health: A Comprehensive Study. Journal of Network and Computer Applications.

Reyes, D. (2021). *Smart contracts and blockchain in real estate transactions: The future of real estate business*. Springer.

Smith, J. (2023). The Impact of Technology on Academic Validation: A Study of Fake Certifications. Academic Press.

Smolenski, N. (2021). Blockchain for Education: A New Credentialing Ecosystem. Frontiers with Artificial Intelligence, Blockchain.

Techopedia. (2024). Top 7 Blockchain Trends 2024: From Institutions to AI. Retrieved from [Techopedia](https://www.techopedia.com) on August 29, 2024.

Truffle Suite. (2020). Truffle - A development framework for Ethereum. Retrieved from https://trufflesuite.com/truffle

Ubaka-Okoye, M., Azeta, A. A., Oni, A. A., Okagbue, H. I., & Nicholas Omoregbe, O. S. (2020). Securing educational data using agent-based blockchain technology. Int. J. Sci. Technol. Res.

Wanyonyi, E. J. (2019). Academic document authenticity verification using blockchain technology (Bachelor's thesis, School of Computing and Informatics).

APPENDIX

Source code available on request.